COFINAL WEIL QUOTIENTS: EXACT TRANSPORT, FIXED-CUTOFF SUMMABILITY, AND THE GLOBAL BRANCH OBSTRUCTION

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Abstract. Connes–Consani–Moscovici construct finite self-adjoint operators from principal Fourier restrictions of the semilocal Weil form, after shifting by the least Ritz value and quotienting by the resulting radical. Suzuki develops a related continuous localized operator and formulates an expected strong-resolvent limit. Starting from these two operator programs, we develop the analytic chain from finite radical quotients to a cutoff-free spectral-gap formulation of Weil positivity.

At a fixed physical cutoff L, we construct canonical isometries between successive positive quotients. The associated generator defect is computed exactly. Before the metric correction it is a sum of two rank-one channels of opposite parity; the correction is the double-operator-integral transform of a rank-two commutator. We give both a divided-difference formula and an explicit Loewner integral, then conjugate the calculation to the natural Euclidean self-adjoint coordinates. This yields exact resolvent and new-mode-shell identities.

For the scalar second-resolvent trace we prove the characteristic formula

$$\operatorname{Tr}(s - D_N)^{-2} = \sum_{|k| \le N} (s - 2\pi k/L)^{-2} - \big(\log \mathcal{C}_N(s)\big)'',$$

where $\mathcal{C}_N$ is the Cauchy transform of the least Ritz vector. It implies absolute local-uniform summability along a cofinal subsequence at every fixed L. We also prove a uniform Weyl bound and show that summability of the transported resolvent defect automatically implies summability of the two-dimensional shell of new modes. For consecutive cutoffs we isolate one sufficient gap–drop series and exhibit a compact-form counterexample showing that abstract Galerkin compactness alone cannot prove that series. We identify the fixed-L sum with the logarithmic curvature of the selected continuum ground transform. After an explicit exterior mesh error $O_K(L^2/N)$, the global limit is equivalent to

$$\partial_z^2 \log(\widehat{\phi}_L/\widehat{k}_L) \longrightarrow 0,$$

where k_L is the CCM prolate model.

We then analyze this global gate. Logarithmic-curvature convergence is proved equivalent, modulo affine normalization, to transform convergence, and a contour formula shows that it carries the complete zero-divisor content. Failure of RH is shown to produce fixed compactly supported negative tests in both parity sectors. Replacing the fixed model by a moving co-Poisson radical yields an exact identity retaining primes, prime powers, Gamma and pole jointly, and a constrained prolate ladder with levels d_4 and d_8, $d_4/d_8 \asymp \lambda^{-8}$.

The branch choice is first isolated as a signed source estimate. On each principal-angle plane the trace projection has eigenvalues $\{-\chi, +\chi\}$: concentration determines its square but not its sign. A translated Montgomery–Vaughan estimate reaches the required $O(\log \lambda)$ prime scale on each Fourier block and the correct local $O(\lambda^2)$ phase-space budget, but loses the block center and does not select the global branch. We then recover that center by an exact identity: the actual prime-power–Gamma multiplier is a nonnegative full-line jump form minus an explicit scalar $\kappa_N = 4\lambda + o(\lambda)$, and has its unique minimum at $t = 0$. In the even sector the pole is a positive rank-one term. A second exact identity cancels the continuous PNT main measure against the growing pole branch and combines the remaining pole branch with Gamma. The sole unresolved arithmetic distribution is $d(\psi(e^u) - e^u)$, coupled to an explicit locally singular kernel. The full diagonal becomes $QW_L = \mathcal{E}_* - c_* I - \mathcal{A}_\Delta$, with $\mathcal{E}_* \ge 0$ and $c_* > 0$. This gives a precise rank-one spectral/discrepancy form of the branch gate and an unconditional far-center theorem for Fejer packets. An exact nonlocal Picone identity and a positive one-atom countermodel show that generic jump positivity and rank-one perturbation cannot prove the remaining one-sided estimate.

For comparison with the classical Weil criterion [1], we then pass from the moving finite-cutoff branch problem to Riemann's full positive kernel K, normalized by $\widehat{K} = \Xi$. Smooth truncations of K are operator quasimodes with super-exponentially small residual. For every even compactly supported multiplier r, the complete Weil form has the exact cutoff-free representation

$$QW(Kr, Kr) = \mathscr{E}_K(r) - \frac{1}{2} \operatorname{Var}_{\mu_K}(r),$$

where $\mathscr{E}_K$ contains every ordinary von Mangoldt atom and the Gamma jump measure, and μ_K is the probability measure induced by the two polar evaluations. Consequently the classical Weil equivalence takes the sharp Poincare inequality

$$\mathscr{E}_K(r) \ge \frac{1}{2} \operatorname{Var}_{\mu_K}(r)$$

on the even test core. Each individual prime-power atom is load-bearing: deleting even one of them makes this sharp inequality false on compact truncations of an exact Riemann radical. The exact theta dilation also shows why retaining only divisible theta indices is insufficient: the nondivisible indices and the central crossing region carry part of the sharp constant.

This last equivalence is a normalized ground-state rewriting of Weil positivity, not a proof

Contents

1. Introduction and scope

Put

$$\xi(s) = \tfrac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \qquad \Xi(t) = \xi(\tfrac{1}{2}+it).$$

The spectral construction of Connes–Consani–Moscovici (CCM) [3] begins with the semilocal Weil form at a physical cutoff $L = 2\log\lambda$, restricts it to a finite Fourier space, shifts by the least Ritz value, and quotients by the one-dimensional radical. The resulting finite operator is self-adjoint in the positive quotient metric, and its characteristic function is a finite real-rooted approximation.

There are two limiting operations:

$$N \longrightarrow \infty \quad (L \text{ fixed}), \qquad L \longrightarrow \infty. \tag{1.1}$$

The first asks whether the finite radical quotients and their arithmetic generators form a coherent system. The second asks whether the resulting spectral divisor is the divisor of Ξ. The purpose of this paper is to resolve a substantial part of the first question without conflating it with the second.

The main conclusions are:

(1) there is a canonical isometric connection between successive shifted Weil quotients;
(2) its arithmetic intertwining defect admits an exact rank-two/Loewner formula;
(3) scalar second-resolvent traces are absolutely summable along a cofinal subsequence for every fixed L;
(4) the apparent new-mode shell is not an independent summability obligation;
(5) consecutive absolute summability requires source regularity not provided by abstract compactness;
(6) logarithmic curvature is an exact relative-divisor observable;
(7) failure of RH is detected by fixed tests in both parity sectors;
(8) a moving co-Poisson vector gives exact coupled prime–Gamma–pole cancellation and a d_4/d_8 prolate hierarchy;
(9) the actual prime–Gamma multiplier has an exact centered jump-square decomposition with unique Fourier center $t = 0$;
(10) after exact PNT compensation, the missing global input is a one-sided estimate for one explicit compensated Stieltjes functional, not three separate source bounds or another absolute norm estimate.
(11) a zero-extension Picone identity is exact but lossless, and a positive one-atom countermodel rules out generic jump positivity as the source of the branch sign.
(12) smooth truncations of Riemann's full kernel are genuine operator quasimodes, not only scalar near-radicals;
(13) ordinary L^2-cyclicity cannot transfer that quasimode estimate to the complementary floor at the required quantitative scale;
(14) the cutoff-free completed Weil form is exactly a full-kernel Doob energy minus one half of an explicit polar variance;
(15) every literal von Mangoldt atom, every nondivisible theta remainder and the central crossing region are load-bearing at that sharp constant.

All finite-level statements below are conditional only on the same simple-even and nonzero-source-overlap hypotheses needed to define the CCM radical quotient in the indicated coordinate. No location hypothesis on the zeros of ζ is used.

2. Starting point in the CCM–Suzuki operator program

CCM construct the finite Weil matrices, identify their rank-two commutator with the periodic derivative, shift by the least eigenvalue, quotient by the radical, and construct the finite rank-one perturbation; see [3, Section 5, especially Lemma 5.4 and Theorem 5.10]. Their two limiting parameters provide the finite spectral architecture used here: first the Fourier dimension tends to infinity at fixed physical cutoff, then the physical cutoff tends to infinity.

Suzuki constructs a continuous localized operator from the Weil distribution and studies its finite-cutoff structure [4]. His continuous screw-function coordinate and expected strong-resolvent limit provide the second starting point. Both programs seek a limiting self-adjoint realization of the zero divisor; neither treats convergence as an automatic consequence of the finite constructions.

The present paper builds the intermediate analytic layers needed to move in that direction. It first constructs a metric-compatible transport between successive CCM quotients, computes its defect and proves fixed-cutoff summability. It then identifies the outer logarithmic-curvature gate, tracks the two branch signs, and keeps primes, prime powers, Gamma and pole coupled through the source compensation. Finally it removes the cutoff from the algebra by using Riemann's full positive kernel as an exact Weil radical. This last step converts the same completed Weil form into a sharp Doob–variance problem.

The Doob coordinate is not asserted to have been the stated endpoint of either source paper. It is a direct ground-state coordinate for the Weil form that underlies both operator programs. It therefore separates two questions which should not be conflated: convergence of the particular finite operators to the zero divisor, and direct positivity of the limiting completed Weil form. The first retains the moving/model identification; the second is equivalent to the sharp full-kernel inequality proved below to be the sole remaining condition in that coordinate.

3. Finite Weil data and the canonical connection

Fix $L > 0$, put $c_L = 2\pi/L$, and let

$$V_N = \operatorname{span}\{U_k : |k| \le N\}, \qquad D_{0,N} U_k = k U_k. \tag{3.1}$$

Let $I_N : V_N \to V_{N+1}$ be Fourier inclusion. The exact principal restriction property of the finite Weil matrices is

$$W_N = I_N^* W_{N+1} I_N. \tag{3.2}$$

Their entries contain the coupled polar, Gamma and prime-power terms of the semilocal Weil form.

Assume that the least eigenvalue is simple, its eigenvector is even, and its overlap with $\eta_N = \sum_{|k|\le N} U_k$ is nonzero. Normalize the ground vector by

$$\varepsilon_N = \min \operatorname{Spec} W_N, \qquad T_N = W_N - \varepsilon_N \mathrm{I}, \qquad \langle \eta_N, \xi_N \rangle = 1. \tag{3.3}$$

Define

$$K_N = (\mathbb{C}\xi_N)^\perp, \quad P_N = P_{K_N}, \quad S_N = T_N|_{K_N}, \quad a_N = D_{0,N}\xi_N, \quad b_N = P_N \eta_N. \tag{3.4}$$

The physical quotient operator represented on K_N is

$$D_N = c_L\big(P_N D_{0,N} P_N - |a_N\rangle\langle b_N|\big), \qquad S_N D_N = D_N^* S_N. \tag{3.5}$$

Set

$$\Delta_N = \varepsilon_N - \varepsilon_{N+1} \ge 0, \qquad C_N = P_{N+1} I_N|_{K_N}. \tag{3.6}$$

Lemma 3.1 (Shifted compression)**.** *For $x, y \in K_N$,*

$$\langle S_{N+1} C_N x, C_N y \rangle = \langle (S_N + \Delta_N \mathrm{I})x, y \rangle. \tag{3.7}$$

Proof. Equation (3.2) gives, for $x, y \in V_N$,

$$\langle T_{N+1} I_N x, I_N y\rangle = \langle T_N x, y\rangle + \Delta_N \langle x, y\rangle.$$

Projecting $I_N x$ and $I_N y$ away from $\ker T_{N+1}$ does not alter the left side, which proves (3.7). $\square$

Put

$$B_N = (S_N + \Delta_N \mathrm{I})^{-1/2} S_N^{1/2}, \qquad j_N = C_N B_N. \tag{3.8}$$

Theorem 3.2 (Canonical radical connection)**.** *The map $j_N : K_N \to K_{N+1}$ is injective and*

$$j_N^* S_{N+1} j_N = S_N. \tag{3.9}$$

Thus it is an isometry for the positive shifted-Weil quotient metrics.

Proof. The right side of (3.7) is positive for $x \neq 0$, so C_N is injective. Substitution of (3.8) into (3.7), using functional calculus for the commuting functions of S_N, gives (3.9). $\square$

The uncorrected inclusion cannot itself descend to the quotient unless $\Delta_N = 0$: if $T_N \xi_N = 0$, then $\|I_N \xi_N\|^2_{T_{N+1}} = \Delta_N \|\xi_N\|^2$. Formula (3.8) is exactly the correction of this obstruction.

4. Exact cross-level generator defect

Define

$$e_N = I_N \xi_N - \xi_{N+1}, \qquad u_N = D_{0,N+1} e_N, \qquad v_N = P_{N+1} I_N \xi_N = P_{N+1} e_N. \tag{4.1}$$

Theorem 4.1 (Two-channel uncorrected defect)**.** *On K_N,*

$$\boxed{X_N := D_{N+1} C_N - C_N D_N = c_L |u_N\rangle\langle b_N| + \frac{c_L}{\|\xi_N\|^2} |v_N\rangle\langle a_N|.} \tag{4.2}$$

The first channel maps even vectors to odd vectors and the second maps odd vectors to even vectors. Hence their nonzero singular values separate:

$$c_L \|u_N\|\,\|b_N\|, \qquad \frac{c_L \|v_N\|\,\|a_N\|}{\|\xi_N\|^2}. \tag{4.3}$$

Moreover,

$$\boxed{\|S_{N+1}^{1/2} v_N\|^2 = \Delta_N \|\xi_N\|^2.} \tag{4.4}$$

Proof. Write $A'_N = D_{0,N} - |a_N\rangle\langle \eta_N|$, so $A'_N \xi_N = 0$. For $x \in K_N$, projection away from the new ground line in the argument of A'_{N+1} may be omitted. Add and subtract $I_N A'_N x$:

$$\begin{aligned} X_N x / c_L &= P_{N+1}(A'_{N+1} I_N - I_N A'_N) x \\ &\quad + P_{N+1} I_N (\mathrm{I} - P_N) A'_N x. \end{aligned}$$

The first line is $|u_N\rangle\langle b_N| x$. Since evenness makes a_N odd and $(A'_N)^* \xi_N = a_N$, the second is $\|\xi_N\|^{-2} |v_N\rangle\langle a_N| x$. Orthogonality of the parity sectors proves (4.3).

For (4.4), apply the shifted compression identity before restricting to K_N to ξ_N, and note that replacing $I_N \xi_N$ by v_N changes it by a vector in $\ker T_{N+1}$. $\square$

5. The metric correction as a Loewner transform

From (3.5),

$$D_N - D_N^* = c_L\big(|b_N\rangle\langle a_N| - |a_N\rangle\langle b_N|\big), \tag{5.1}$$

and metric self-adjointness gives

$$\boxed{[D_N, S_N] = c_L\big(|b_N\rangle\langle S_N a_N| - |a_N\rangle\langle S_N b_N|\big).} \tag{5.2}$$

This commutator has rank at most two.

Let

$$f_\Delta(t) = \sqrt{\frac{t}{t+\Delta}}, \qquad B_N = f_{\Delta_N}(S_N). \tag{5.3}$$

If $S_N = \sum_p \sigma_p E_p$, the finite-dimensional double-operator- integral identity is

$$\boxed{[D_N, B_N] = \sum_{p,q} f^{[1]}_{\Delta_N}(\sigma_p, \sigma_q) E_p [D_N, S_N] E_q,} \tag{5.4}$$

where $f^{[1]}(x, y) = (f(y) - f(x))/(y - x)$, with the derivative on the diagonal. The representation

$$f_\Delta(t) = \frac{1}{\pi}\int_0^1 \frac{t}{t + r\Delta}\frac{dr}{\sqrt{r(1-r)}} \tag{5.5}$$

and (5.2) give the more explicit formula

$$\boxed{\begin{aligned}[D_N, B_N] = \frac{c_L\Delta_N}{\pi}\int_0^1 \sqrt{\frac{r}{1-r}}\big(&|R_r b_N\rangle\langle S_N R_r a_N| \\ &- |R_r a_N\rangle\langle S_N R_r b_N|\big)\, dr,\end{aligned}} \tag{5.6}$$

with $R_r = (S_N + r\Delta_N \mathrm{I})^{-1}$.

Since f_Δ is operator monotone, its Loewner matrix is positive. If $m_N = \min \operatorname{Spec} S_N$, the induced completely positive Schur multiplier gives, for every $1 \le p \le \infty$,

$$\boxed{\|[D_N, B_N]\|_{\mathcal{S}_p} \le \frac{\Delta_N}{2\sqrt{m_N}(m_N + \Delta_N)^{3/2}}\|[D_N, S_N]\|_{\mathcal{S}_p}.} \tag{5.7}$$

Theorem 5.1 (Complete transport defect)**.** *The arithmetic intertwining defect $\mathcal{R}_N = D_{N+1} j_N - j_N D_N$ is*

$$\boxed{\mathcal{R}_N = X_N B_N + C_N [D_N, B_N].} \tag{5.8}$$

Consequently it is determined explicitly by (4.2) and (5.6), with no input from the zeros of Ξ.

Proof. Use $j_N = C_N B_N$, add and subtract $C_N D_N B_N$, and apply (4.2). ☐

Raw Euclidean norms of $\mathcal{R}_N$ do not measure the canonical transport. Put

$$\mathsf{A}_N = S_N^{1/2} D_N S_N^{-1/2}, \qquad \mathsf{U}_N = S_{N+1}^{1/2} C_N (S_N + \Delta_N \mathrm{I})^{-1/2}. \tag{5.9}$$

Then A_N is Euclidean self-adjoint and U_N is an isometry. The correct defect is

$$\widehat{\mathcal{R}}_N = \mathsf{A}_{N+1}\mathsf{U}_N - \mathsf{U}_N \mathsf{A}_N = S_{N+1}^{1/2} \mathcal{R}_N S_N^{-1/2}. \tag{5.10}$$

Writing $H_N = S_N + \Delta_N \mathrm{I}$,

$$\boxed{\widehat{\mathcal{R}}_N = S_{N+1}^{1/2} X_N H_N^{-1/2} + \mathsf{U}_N Y_N,} \tag{5.11}$$

where

$$Y_N = \frac{c_L \Delta_N}{\pi} \int_0^1 \sqrt{\frac{r}{1-r}} \big(|H_N^{1/2} R_r b_N\rangle\langle S_N^{1/2} R_r a_N| \\ - |H_N^{1/2} R_r a_N\rangle\langle S_N^{1/2} R_r b_N| \big)\, dr. \tag{5.12}$$

The two singular values of the first term in (5.11) are

$$\alpha_N = c_L \|S_{N+1}^{1/2} u_N\| \, \|H_N^{-1/2} b_N\|, \quad \beta_N = \frac{c_L \sqrt{\Delta_N}}{\|\xi_N\|} \|H_N^{-1/2} a_N\|. \tag{5.13}$$

6. Resolvent transport and the two-mode shell

For $z \notin \mathbb{R}$, let

$$G_N(z) = (\mathsf{A}_N - z)^{-1}, \qquad E_N = G_{N+1} \mathsf{U}_N - \mathsf{U}_N G_N. \tag{6.1}$$

The resolvent identity gives

$$\boxed{E_N = -G_{N+1} \widehat{\mathcal{R}}_N G_N.} \tag{6.2}$$

Let

$$Q_N^{\mathrm{new}} = \mathrm{I} - \mathsf{U}_N \mathsf{U}_N^*. \tag{6.3}$$

The dimension of its range is two, one mode in each parity sector.

Proposition 6.1 (Exact second-resolvent increment)**.**

$$\boxed{\begin{aligned} \operatorname{Tr} G_{N+1}(z)^2 - \operatorname{Tr} G_N(z)^2 = {} & \operatorname{Tr} \mathsf{U}_N^* (G_{N+1} E_N + E_N G_N) \\ & + \operatorname{Tr} Q_N^{\mathrm{new}} G_{N+1}(z)^2 Q_N^{\mathrm{new}}. \end{aligned}} \tag{6.4}$$

Proof. Insert $\mathrm{I} = \mathsf{U}_N \mathsf{U}_N^* + Q_N^{\mathrm{new}}$ into the trace of G_{N+1}^2. In the transported block use $G_{N+1}\mathsf{U}_N = \mathsf{U}_N G_N + E_N$, expand once from each side, and use cyclicity of the finite trace. ☐

There is also an exact Hilbert–Schmidt identity. Put $H_N(z) = \|G_N(z)\|_{\mathcal{S}_2}^2$. Then

$$\boxed{\|G_{N+1} Q_N^{\mathrm{new}}\|_{\mathcal{S}_2}^2 = H_{N+1} - H_N - 2\Re \operatorname{Tr}((\mathsf{U}_N G_N)^* E_N) - \|E_N\|_{\mathcal{S}_2}^2.} \tag{6.5}$$

7. The scalar characteristic and fixed-L summability

Let ϕ_N be an L^2-unit least Ritz vector of W_N. Define

$$\mathcal{C}_N(s) = \sum_{k=-N}^{N} \frac{(\phi_N)_k}{s - c_L k}, \qquad s \in \mathbb{C} \setminus \mathbb{R}, \tag{7.1}$$

and

$$\tau_N(s) = \operatorname{Tr}(s - D_N)^{-2}. \tag{7.2}$$

Theorem 7.1 (Exact Cauchy-transform formula)**.**

$$\boxed{\tau_N(s) = \sum_{k=-N}^{N} \frac{1}{(s - c_L k)^2} - \big(\log \mathcal{C}_N(s)\big)''.} \tag{7.3}$$

Proof. First take the dimensionless diagonal $D_0 e_k = k e_k$, normalize ξ_N by $\langle \eta_N, \xi_N \rangle = 1$, and write $D'_N = D_0 - |D_0 \xi_N\rangle\langle \eta_N|$. The matrix determinant lemma gives

$$\begin{aligned} \det(s - D'_N) &= \prod_{k=-N}^{N} (s-k) \left(1 + \sum_{k=-N}^{N} \frac{k(\xi_N)_k}{s-k}\right) \\ &= s \prod_{k=-N}^{N} (s-k) \sum_{k=-N}^{N} \frac{(\xi_N)_k}{s-k}. \end{aligned} \tag{7.4}$$

The factor s is the radical eigenvalue and is removed on the quotient. Rescale $s \mapsto s/c_L$; multiplication of the ground vector by a nonzero scalar disappears after two logarithmic derivatives. Finally use $\mathrm{Tr}(s-D)^{-2} = -(\log \det(s-D))''$. □

Finite self-adjointness implies that $\mathcal{C}_N$ is zero-free off the real axis.

7.1. Quotient trace versus the full regularized determinant.

7.1. Quotient trace versus the full regularized determinant. Formula (7.3) is the trace of the finite quotient. The full CCM regularized determinant also retains the free exterior Fourier mesh. In the present affine coordinate its second-logarithmic-derivative contribution is

$$T^{\mathrm{ext}}_{L,N}(s) = \sum_{|k|>N} \frac{1}{(s - c_L k)^2}. \tag{7.5}$$

Thus, up to the sign convention used for logarithmic curvature and an affine regularization phase (which disappears after two derivatives), the full trace is $\tau_{L,N} + T^{\mathrm{ext}}_{L,N}$. If $K \Subset \mathbb{C} \setminus \mathbb{R}$ and N is large enough relative to L and K, then

$$\sup_{s \in K} |T^{\mathrm{ext}}_{L,N}(s)| \le C_K \frac{L^2}{N}. \tag{7.6}$$

Indeed, $|s - c_L k| \gg_K |k|/L$ on the exterior tail and $\sum_{k>N} k^{-2} = O(N^{-1})$. Therefore fixed-$L$ passage to $N \to \infty$ is harmless. On a diagonal limit, however, discarding the exterior factor requires

$$\frac{N(L)}{L^2} \longrightarrow \infty. \tag{7.7}$$

No such rate is required if the known exterior factor is retained or divided out exactly. This normalization distinction is essential: the finite quotient trace and the full regularized determinant are not the same object before the exterior mesh is accounted for.

7.2. A sufficient consecutive rate. Let

$$g_{N+1} = \lambda_2(W_{N+1}) - \varepsilon_{N+1} > 0, \tag{7.8}$$

and choose the phases so that $\langle I_N \phi_N, \phi_{N+1} \rangle \ge 0$.

Lemma 7.2 (Ground-line rotation)**.**

$$\|I_N \phi_N - \phi_{N+1}\|^2 \le \frac{2\Delta_N}{g_{N+1}}. \tag{7.9}$$

Proof. Write $I_N\phi_N = a\phi_{N+1} + h$, with $a \geq 0$ and $h \perp \phi_{N+1}$. The min–max principle yields $\Delta_N \geq g_{N+1}\|h\|^2$. Since $a^2 + \|h\|^2 = 1$ and $1 - a \leq 1 - a^2$, one has $\|I_N\phi_N - \phi_{N+1}\|^2 = 2(1-a) \leq 2\|h\|^2$. □

Theorem 7.3 (Consecutive scalar criterion). *If*

$$\sum_{N \geq N_0} \sqrt{\Delta_N / g_{N+1}} < \infty, \tag{7.10}$$

then, for every compact $K \Subset \mathbb{C} \setminus \mathbb{R}$,

$$\sum_{N \geq N_0} \sup_{s \in K} |\tau_{N+1}(s) - \tau_N(s)| < \infty. \tag{7.11}$$

Proof. For $j = 0, 1, 2$, the vectors

$$\left(\frac{d^j}{ds^j} \frac{1}{s - c_L k} \right)_{k \in \mathbb{Z}}$$

belong uniformly to $\ell^2(\mathbb{Z})$ for $s \in K$. Equations (7.9)–(7.10) imply $\sum_N \|\mathcal{C}_{N+1} - \mathcal{C}_N\|_{C^2(K)} < \infty$. The limit is nonzero and, by Hurwitz, zero-free off $\mathbb{R}$. After finitely many indices, $f \mapsto (\log f)''$ is Lipschitz on this bounded subset of $C^2(K)$. The two new lattice summands in (7.3) are $O_K(N^{-2})$. Summing proves (7.11). □

7.3. **Rate-free cofinal summability.**

Theorem 7.4 (Fixed-L cofinal summability). *Assume that the quotient construction is defined on a cofinal set of levels. There exists a cofinal sequence $N_j \to \infty$ such that, for every compact $K \Subset \mathbb{C} \setminus \mathbb{R}$,*

$$\boxed{\sum_j \sup_{s \in K} |\tau_{N_{j+1}}(s) - \tau_{N_j}(s)| < \infty.} \tag{7.12}$$

The same sequence can be chosen simultaneously for a countable exhaustion of $\mathbb{C} \setminus \mathbb{R}$.

Proof. The unit Ritz vectors have uniformly bounded shifted form norm because their Rayleigh values decrease to the lower bound of the closed semilocal Weil form. The compact form embedding proved in the CCM construction makes them precompact in L^2. Select coherent phases and a subsequence with

$$\phi_{N_j} \to \phi, \qquad \|\phi_{N_j} - \phi\|_2 \leq 2^{-j-2}. \tag{7.13}$$

Lower semicontinuity shows that ϕ is a continuum ground state and has norm one. Hence $\sum_j \|\phi_{N_{j+1}} - \phi_{N_j}\|_2 < \infty$.

The ℓ^2 estimate used above now gives bounded variation of $\mathcal{C}_{N_j}$ in $C^2(K)$. Its limit cannot vanish identically, because the Cauchy transform determines all Fourier coefficients of the nonzero vector ϕ. Hurwitz gives zero-freeness in both half-planes. Finally, the lattice shells

$$\sum_{N_j < |k| \leq N_{j+1}} (s - c_L k)^{-2}$$

are disjoint and absolutely summable, uniformly on K. Formula (7.3) proves (7.12). A diagonal selection over a countable compact exhaustion gives the last assertion. □

Corollary 7.5 (Selected continuum transform). *The Fourier transform of the continuum ground state selected in the proof has no nonreal zero.*

Proof. Compact support and L^2-convergence imply locally uniform convergence of the Fourier transforms. Each finite transform has only real zeros by the CCM determinant theorem. The limit is not identically zero, so Hurwitz excludes nonreal zeros. □

Theorem 7.6 (Identification of the fixed-L sum)**.** *Let ϕ_L be the continuum ground state selected above and write its Fourier transform in the CCM normalization as*

$$\widehat{\phi}_{L,N}(z) = 2L^{-1/2}\sin\left(\frac{Lz}{2}\right)\sum_{k=-N}^{N}\frac{(\phi_{L,N})_k}{z-c_Lk}. \tag{7.14}$$

Then the limit in (7.12) is not merely an abstract holomorphic function:

$$\boxed{\tau_L(z) := \lim_{j\to\infty}\tau_{L,N_j}(z) = -\partial_z^2\log\widehat{\phi}_L(z)} \tag{7.15}$$

locally uniformly on $\mathbb{C}\setminus\mathbb{R}$.

Proof. Compact support turns L^2-convergence of the selected ground vectors into locally uniform convergence of their Fourier transforms with all derivatives. The full regularized determinant is, up to a nonzero scalar,

$$F_{L,N}(z) = -ie^{-iLz/2}\widehat{\phi}_{L,N}(z).$$

The exponential is affine after taking a logarithm, hence invisible to the second derivative. The full and quotient traces differ by the exterior tail (7.5), which tends locally uniformly to zero at fixed L by (7.6). Taking the limit in $(\log F_{L,N})'' = -\tau_{L,N}^{\mathrm{full}}$ proves (7.15). □

8. Uniform Weyl control and elimination of the shell

Theorem 8.1 (Uniform fixed-L Weyl bound)**.** *For each $z\notin\mathbb{R}$,*

$$\boxed{\sup_N\|G_N(z)\|_{\mathcal{S}_2}^2<\infty.} \tag{8.1}$$

Proof. Let $\mathcal{K}_L$ be the L^2-closure of the unit Ritz vectors. The compact form embedding makes $\mathcal{K}_L$ compact and disjoint from zero. Fourier transform is continuous and injective on this set. Hence there exist finitely many points $w_1,\dots,w_r$ and $c>0$ such that

$$\max_j|\widehat{f}(w_j)|\ge c \qquad (f\in\mathcal{K}_L). \tag{8.2}$$

After translating the support to $[-L/2,L/2]$,

$$|\widehat{f}(w)|\le\sqrt{L}\,e^{(L/2)|\Im w|}. \tag{8.3}$$

Jensen's formula centered at a point satisfying (8.2), together with (8.3), gives the uniform zero count

$$\#\{\theta\in\operatorname{Spec}\mathsf{A}_N:|\theta|\le R\}\le C_L(1+R). \tag{8.4}$$

Here multiplicity is retained through the finite characteristic identity. Dyadic summation of (8.4) yields

$$\sup_N\sum_{\theta\in\operatorname{Spec}\mathsf{A}_N}|\theta-z|^{-2}<\infty,$$

which is (8.1). □

Corollary 8.2 (The new shell is not independent)**.** *If*

$$\sum_N \|E_N\|_{\mathcal{S}_1} = \sum_N \|G_{N+1}\widehat{\mathcal{R}}_N G_N\|_{\mathcal{S}_1} < \infty, \tag{8.5}$$

then

$$\boxed{\sum_N \|G_{N+1}Q_N^{\mathrm{new}}\|_{\mathcal{S}_2}^2 < \infty.} \tag{8.6}$$

Proof. Since $\|\mathsf{U}_N G_N\| \le |\Im z|^{-1}$,

$$|\operatorname{Tr}((\mathsf{U}_N G_N)^* E_N)| \le |\Im z|^{-1}\|E_N\|_{\mathcal{S}_1}.$$

Hypothesis (8.5) also implies $\sum_N \|E_N\|_{\mathcal{S}_2}^2 < \infty$. Sum the nonnegative left side of (6.5) and use the uniform bound (8.1). □

Thus the two-series trace-Cauchy package reduces to the single transported series (8.5).

9. Why compactness does not prove consecutive summability

CCM's monotone Ritz theorem gives

$$\sum_N \Delta_N = \varepsilon_{N_0} - \inf \operatorname{Spec} A_L < \infty, \tag{9.1}$$

but this does not imply (7.10), even when the limiting spectral gap is positive.

Proposition 9.1 (Compact-form counterexample)**.** *There is a closed positive form with compact form embedding, a simple ground state, a positive limiting gap and nested coordinate cores for which*

$$\Delta_N \asymp N^{-2}, \qquad g_N \ge g > 0, \qquad \sum_N \sqrt{\Delta_N/g_N} = \infty. \tag{9.2}$$

Proof. On $\mathcal{H} = \mathbb{C} \oplus \ell^2(\mathbb{N})$, define

$$q(x) = \sum_{n\ge 1} n\,|x_n - n^{-3/2}x_0|^2. \tag{9.3}$$

The x_0-direction is finite-dimensional, and on $x_0 = 0$ the form controls $\sum n|x_n|^2$; hence its form embedding is compact. Its kernel is the simple line generated by $(1, n^{-3/2})$, and compactness gives a positive gap above it.

Restrict to coordinates $0, 1, \ldots, N$. Eliminating the last N coordinates in the least-eigenvalue equations gives exactly

$$t_N = \varepsilon_N\left(1 + \sum_{n\le N}\frac{1}{n^2(n-\varepsilon_N)}\right), \qquad t_N = \sum_{n>N}\frac{1}{n^2}. \tag{9.4}$$

Therefore

$$\varepsilon_N \sim \frac{1}{(1+\zeta(3))N}, \qquad \Delta_N \sim \frac{1}{(1+\zeta(3))N^2}, \tag{9.5}$$

which proves (9.2). □

Consequently, consecutive summability must use additional regularity of the actual polar–Gamma–Euler ground state. For example, a source-specific Ritz estimate

$$\varepsilon_N - \inf \operatorname{Spec} A_L = O(N^{-p} \log N), \qquad p > 1, \tag{9.6}$$

together with a limiting positive gap would suffice. Neither audited source proves such a rate.

10. Reproducible finite-level check

The accompanying NumPy-only script `106_04_cofinal_weil_defect_probe.py`, retained in the research archive, constructs one largest Weil matrix and obtains all lower levels by principal compression. It checks the shifted compression, the rank-two formula (4.2), the complete defect (5.8), the metric conjugation (5.10), the resolvent identity (6.2), and the shell increment (6.4). A separate scalar script checks (7.3) and the summability bookkeeping. These calculations are diagnostic: floating-point decay is not used in any theorem.

11. The remaining global limit

Put

$$s = \frac{1}{2} + iz, \qquad \Xi(z) = \xi\left(\frac{1}{2} + iz\right). \tag{11.1}$$

The chain rule gives the sign-correct identity

$$\boxed{\partial_z^2 \log \Xi(z) = -\left(\frac{\xi'}{\xi}\right)'\left(\frac{1}{2} + iz\right).} \tag{11.2}$$

If $\Theta_{L,N}^{\mathrm{full}} = \frac{1}{2} + iD_{L,N}^{\mathrm{full}}$, then

$$\operatorname{Tr}(s - \Theta_{L,N}^{\mathrm{full}})^{-2} = -\tau_{L,N}^{\mathrm{full}}(z). \tag{11.3}$$

Thus the desired global statement can be written equivalently as

$$\begin{aligned} \operatorname{Tr}(s - \Theta_{L,N(L)}^{\mathrm{full}})^{-2} &\longrightarrow -\left(\frac{\xi'}{\xi}\right)'(s), \\ \tau_{L,N(L)}^{\mathrm{full}}(z) &\longrightarrow \left(\frac{\xi'}{\xi}\right)'\left(\frac{1}{2} + iz\right). \end{aligned} \tag{11.4}$$

There is no additional sign or scale factor.

Let k_L be the explicit prolate model denoted k_λ by CCM, where $L = 2\log\lambda$. Their model theorem gives

$$\widehat{k}_L \longrightarrow \Xi \tag{11.5}$$

locally uniformly on closed substrips of $|\Im z| < 1/2$. On compact sets which avoid the zeros of Ξ, Cauchy's theorem therefore gives

$$\partial_z^2 \log \widehat{k}_L \longrightarrow \partial_z^2 \log \Xi. \tag{11.6}$$

Combining (7.15) and (11.6) yields an exact affine-free reduction.

Theorem 11.1 (Affine-free ground/model criterion)**.** *Assume that a simple-even continuum ground state ϕ_L is selected for cofinally many L. Choose the finite diagonal $N = N(L)$ from the fixed-L cofinal sequences so that, on a compact exhaustion,*

$$\tau_{L,N(L)}^{\mathrm{q}} - \tau_L \longrightarrow 0 \quad \textit{locally uniformly.} \tag{11.7}$$

Such a diagonal exists by (7.12). If the exterior factor is discarded, choose it simultaneously so that $N(L)/L^2 \to \infty$; if that factor is retained or divided out exactly, no exterior rate is needed. Under these diagonal conditions, the global second-resolvent identification (11.4) holds if and only if

$$\boxed{\partial_z^2 \log\left(\frac{\widehat{\phi}_L(z)}{\widehat{k}_L(z)}\right) \longrightarrow 0} \tag{11.8}$$

locally uniformly on every simply connected compact set in the open strip which avoids the zeros of both transforms.

Proof. Choose compatible logarithms on the compact set. The identity

$$\partial_z^2 \log \widehat{\phi}_L - \partial_z^2 \log \Xi = \partial_z^2 \log \frac{\widehat{\phi}_L}{\widehat{k}_L} + \partial_z^2 \log \frac{\widehat{k}_L}{\Xi} \tag{11.9}$$

is exact. Its last term tends to zero by (11.5) and Cauchy estimates. Now use (7.15) and (11.2)–(11.4). The finite-diagonal assertion follows from the exterior bound (7.6). $\square$

The curvature criterion is invariant under $\widehat{\phi}_L(z) \mapsto e^{a_L + b_L z}\widehat{\phi}_L(z)$; hence it removes the full admissible affine normalization.

Lemma 11.2 (Quantitative relative-error transfer)**.** *Let K be compact and let K_r be its closed r-neighborhood inside a zero-free domain. Put*

$$m_{L,K,r} = \inf_{z\in K_r} |\widehat{k}_L(z)|, \qquad \eta_{L,K,r} = \sup_{z\in K_r} |c_L^*\widehat{\phi}_L(z) - \widehat{k}_L(z)|. \tag{11.10}$$

If $\eta_{L,K,r} \le m_{L,K,r}/2$, then

$$\boxed{\sup_{z\in K}\left|\partial_z^2 \log \frac{c_L^*\widehat{\phi}_L(z)}{\widehat{k}_L(z)}\right| \le \frac{4}{r^2}\frac{\eta_{L,K,r}}{m_{L,K,r}}.} \tag{11.11}$$

Here $c_L^* \neq 0$ is any source normalization.

Proof. Write $c_L^*\widehat{\phi}_L/\widehat{k}_L = 1+h_L$. On K_r, $|h_L| \le 1/2$ and $|\log(1+h_L)| \le 2|h_L|$. Cauchy's estimate for the second derivative on disks of radius r gives (11.11). $\square$

A stronger source-side sufficient condition is

$$\sup_{|\beta|\le B} \int_{\lambda^{-1}}^{\lambda} |c_L^*\phi_L(u) - k_L(u)| u^\beta \, d^*u \longrightarrow 0, \qquad B < \frac{1}{2}. \tag{11.12}$$

Indeed, $|u^{-iz}| = u^{\Im z}$, so (11.12) bounds the numerator in (11.10) on closed substrips. Model convergence keeps the denominator uniformly positive away from the zeros of Ξ, and the lemma implies (11.8).

In the operator program adopted here, (11.8) and (11.12) are the force-bearing global source-identification estimates. They do not follow from fixed-L compactness, and the present paper does not assume them.

12. Curvature rigidity and the zero divisor

The remaining limit is not weakened by taking two logarithmic derivatives. It is merely made insensitive to affine normalization.

For a zero-free holomorphic function F, put

$$\mathcal{K}_F := -(\log F)''.$$

Theorem 12.1 (Affine rigidity)**.** *Let $\Omega \subset \mathbb{C}$ be simply connected, and let F_j, G_j be holomorphic and zero-free on Ω. Then*

$$\mathcal{K}_{F_j} - \mathcal{K}_{G_j} \longrightarrow 0$$

locally uniformly if and only if there are $a_j, b_j \in \mathbb{C}$ such that

$$e^{-a_j - b_j z} \frac{F_j(z)}{G_j(z)} \longrightarrow 1$$

locally uniformly. If $0 \in \Omega$, both functions are even, and $F_j(0) = G_j(0) \neq 0$, then $a_j = b_j = 0$.

Proof. Choose $H_j = \log(F_j/G_j)$, fix $z_0 \in \Omega$, and subtract

$$A_j(z) = H_j(z_0) + (z - z_0)H_j'(z_0).$$

Two integrations along bounded paths in a slightly larger compact set give

$$\sup_K |H_j - A_j| \leq C_K \sup_{K'} |H_j''|.$$

This proves one direction after exponentiation; the converse follows from Cauchy estimates. In the even normalized case, choose $H_j(0) = 0$. Then $H_j'(0) = 0$, so the affine correction vanishes. ∎

Curvature also counts the relative zero divisor directly. If F is holomorphic near the closure of a Jordan domain D, with no zero on $C = \partial D$, then for every $z_* \in \mathbb{C}$,

$$\boxed{N_F(D) = -\frac{1}{2\pi i} \int_C (z - z_*)\mathcal{K}_F(z)\, dz.} \tag{12.1}$$

Indeed, a zero of multiplicity m contributes $\mathcal{K}_F(z) = -m(z-a)^{-2} + O(1)$, and the residue in (12.1) is $-m$.

Theorem 12.2 (Divisor rigidity)**.** *Assume that every selected transform $\widehat{\phi}_L$ has only real zeros, $\widehat{k}_L \to \Xi$ locally uniformly, and*

$$\mathcal{K}_{\widehat{\phi}_L} - \mathcal{K}_{\widehat{k}_L} \longrightarrow 0$$

uniformly on every fixed contour avoiding both divisors. Then every zero of Ξ is real.

Proof. If Ξ had a nonreal zero, choose a small disk D around it, disjoint from the real axis. Rouché's theorem gives $N_{\widehat{k}_L}(D) > 0$ for large L, whereas real-rootedness gives $N_{\widehat{\phi}_L}(D) = 0$. Formula (12.1) and curvature convergence force their difference to tend to zero, a contradiction. ∎

Thus (11.8) already has the full zero-location content of RH. The next sections isolate its exact variational and arithmetic requirements.

13. Variational selection and bilateral detection

Let A_L be the even continuum semilocal Weil operator, with normalized simple ground state ϕ_L, first two eigenvalues $\epsilon_{0,L} < \epsilon_{1,L}$, and gap $g_L = \epsilon_{1,L} - \epsilon_{0,L}$. For a normalized model q_L, put $R_L = \langle A_L q_L, q_L\rangle$. Spectral expansion gives

$$1 - |\langle q_L, \phi_L\rangle|^2 \leq \frac{R_L - \epsilon_{0,L}}{g_L}. \tag{13.1}$$

Consequently, with $L = 2\log\lambda$,

$$\sup_{|\Im z|\leq B} |\widehat{q}_L(z) - e^{i\vartheta_L}\widehat{\phi}_L(z)| \leq W_{L,B}\left(\frac{2(R_L - \epsilon_{0,L})}{g_L}\right)^{1/2}, \tag{13.2}$$

where

$$W_{L,B}^2 = \frac{\lambda^{2B} - \lambda^{-2B}}{2B}, \qquad W_{L,0}^2 = L.$$

Together with (11.11), this proves the curvature limit whenever

$$\boxed{\frac{\|k_L\|_2 W_{L,B}}{\inf|\widehat{k}_L|}\left(\frac{R_L - \epsilon_{0,L}}{g_L}\right)^{1/2} \longrightarrow 0.} \tag{13.3}$$

There is a weaker gate which proves RH without identifying the complete ground transform.

Theorem 13.1 (Residual/overlap criterion). *Let $A_L\phi_L = \epsilon_L\phi_L$, $\|\phi_L\| = \|q_L\| = 1$, and $\mu_L \in \mathbb{R}$. Then*

$$\boxed{|\epsilon_L - \mu_L|\,|\langle\phi_L, q_L\rangle| \leq \|(A_L - \mu_L)q_L\|.} \tag{13.4}$$

Hence $\mu_L \to 0$ and a vanishing residual/overlap quotient force $\epsilon_L \to 0$.

Proof. Take the inner product of $(A_L - \mu_L)q_L$ with ϕ_L, use self-adjointness, and apply Cauchy–Schwarz. □

To use (13.4), failure of RH must be detected in the selected parity. The following strengthens the usual unspecified-parity Weil test.

Theorem 13.2 (Bilateral parity detection). *If RH is false, there exist fixed nonzero real functions*

$$f_+ \in C_c^\infty(\mathbb{R})^{\mathrm{even}}, \qquad f_- \in C_c^\infty(\mathbb{R})^{\mathrm{odd}},$$

such that

$$QW(f_+, f_+) < 0, \qquad QW(f_-, f_-) < 0. \tag{13.5}$$

Consequently both semilocal parity bottoms are bounded above by a fixed negative constant for every sufficiently large support window.

Proof. Use the centered zero parameter s, so that an off-line quartet is $\{s_0, \bar{s}_0, -s_0, -\bar{s}_0\}$, with $\Re s_0 \Im s_0 \neq 0$. For a real even test with transform F, the quartet contributes $4m\Re F(s_0)^2$; for a real odd test it contributes $-4m\Re F(s_0)^2$. Thus the target values $F_+(s_0) = i$ and $F_-(s_0) = 1$ make both contributions negative.

Choose a real even Paley–Wiener damping transform Φ, first fixing it and then a finite zero box on whose complement $|\Phi(s)/\Phi(s_0)| \leq 1/2$. Delete the full target orbit, with multiplicity, and let Z_R be the real even polynomial annihilating the remaining finite divisor. Since $\Im(s_0^2) \neq 0$, the real-linear map $(a,b)\mapsto a+bs_0^2$ is an isomorphism. Hence one can choose real coefficients so that

$$F_{N,+}(z)=Z_R(z)(a_N+b_Nz^2)\Phi(z)^N,$$
$$F_{N,-}(z)=izZ_R(z)(c_N+d_Nz^2)\Phi(z)^N$$

have the prescribed target values. Paley–Wiener gives fixed-parity compactly supported smooth inverse transforms. Rapid strip decay and the zero count $N(T)=O(T\log T)$ dominate the remaining zero sum by a summable majorant times 4^{-N}. For one sufficiently large finite N, both full Weil values are negative. Rayleigh's principle proves the semilocal assertion. □

Combining Theorems 13.1 and 13.2, either parity ground proves RH if

$$\boxed{\frac{\|A_Lq_L\|}{|\langle\phi_L,q_L\rangle|}\longrightarrow 0.} \tag{13.6}$$

This gate is weaker than (13.3), but neither an endpoint normalization nor qualitative positivity can control its denominator.

Remark 13.3 (Sharpness of both quotients). For $0<\eta<1$, the positivity-improving matrix

$$A_\eta=\begin{pmatrix}-1 & -\eta\\ -\eta & 0\end{pmatrix},\qquad q_\eta=\frac{(\eta^2,1)}{\|(\eta^2,1)\|}$$

has a simple strictly positive ground whose eigenvalue tends to -1, while $\|A_\eta q_\eta\|\sim\eta$ and its overlap with the ground is also asymptotic to η. Thus the quotient in (13.6) does not improve.

Likewise, on $L^2([-1,1])^{\mathrm{even}}$, the two real-rooted transforms $\sin z/z$ and $(\sin(z/2)/(z/2))^2$ can be made respectively the ground and a quasimode with Rayleigh excess tending to zero but a gap collapsing at the same rate. Their logarithmic curvatures remain different. Hence compact resolvent, simplicity, real-rootedness and an unscaled quasimode estimate cannot replace (13.3).

14. Near-radical clusters and the model scale

Let K be Riemann's full additive kernel, normalized by $\widehat{K}=\Xi$. Every derivative $K_j=\partial_x^jK$ lies in the polarized Weil radical:

$$QW(K_j,g)=0, \tag{14.1}$$

because $\widehat{K}_j(z)=(iz)^j\Xi(z)$ vanishes on the complete nontrivial divisor.

For $b>1/2$, define

$$\|f\|_{\mathcal{X}_b}=\sum_{r=0}^{2}\sup_x e^{b|x|}(1+|x|)^3|f^{(r)}(x)|.$$

The completed source formula satisfies the unconditional continuity bound

$$\boxed{|QW(f,g)|\le C_b\|f\|_{\mathcal{X}_b}\|g\|_{\mathcal{X}_b}.} \tag{14.2}$$

Indeed, two derivatives control the symmetric Gamma difference at the origin, exponential decay controls its tail and the polar evaluations, and the prime powers are bounded by

$$\sum_{m\ge 2}\frac{\Lambda(m)}{m^{1/2+b'}}=-\frac{\zeta'}{\zeta}\left(\frac12+b'\right)<\infty\qquad(1/2<b'<b).$$

Choose a smooth cutoff χ_a, equal to one on $[-a,a]$, and put $v_{j,a} = \chi_a K_j$, $e_{j,a} = v_{j,a} - K_j$, with $a = \log\lambda$. The theta-series decay of K gives

$$\|e_{j,a}\|_{\mathcal{X}_b} \le C_j \lambda^{M_j} e^{-c_j\lambda^2}.$$

Using (14.1) twice,

$$QW(v_{i,a}, v_{j,a}) = QW(e_{i,a}, e_{j,a}).$$

Therefore, for $i,j \in \{0,1,2\}$,

$$\boxed{|QW(v_{i,a}, v_{j,a})| \le C\lambda^M e^{-c\lambda^2}.} \tag{14.3}$$

Min–max applied to $\mathrm{span}\{v_{0,a}, v_{2,a}\}$ and $\mathrm{span}\{v_{1,a}\}$ gives unconditional upper bounds of this size for the second even and first odd eigenvalues. Under RH, and only then using Weil positivity, the next-even and parity gaps collapse at the same double-exponential-in-L scale. Thus no fixed, polynomial or ordinary exponential gap can close (13.3).

The quantitative CCM model estimate has a different and much coarser scale. With k_λ the compact prolate model and k the full kernel, the Meixner estimate and the Gaussian exterior tail give

$$\boxed{\|k_\lambda - k\|_2^2 = O(\lambda^{-1}), \qquad \sup_{|\Im z| \le B} |\widehat{k}_\lambda(z) - \Xi(z)| = O_B(\lambda^{B-1/2})} \tag{14.4}$$

for every fixed $B < 1/2$.

This does not control the Weil form. The Gamma multiplier grows like $(2\pi)^{-1}\log|t|$, and hence is unbounded in the L^2 topology. The finite prime block satisfies only

$$\left\| \sum_{1<n\le\lambda^2} \Lambda(n) T(n) \right\| \le 4C_\psi \lambda, \tag{14.5}$$

which consumes the full squared gain in (14.4). Although radicality gives the exact scalar identity

$$QW(k_\lambda, k_\lambda) = QW(k_\lambda - k, k_\lambda - k), \tag{14.6}$$

the available model estimate proves neither $R_L \to 0$, nor an operator residual, nor the weighted quotient (13.3).

15. The moving co-Poisson radical

The fixed model error in (14.4) can be removed from the exact identity by moving inside the full co-Poisson radical.

For an even compactly supported BV function f, supported in $[-\lambda,\lambda]$ and satisfying $\int_0^\infty f = 0$, define

$$\mathcal{E}f(u) = u^{1/2} \sum_{n\ge1} f(nu). \tag{15.1}$$

Then $F = \mathcal{E}f \in L^2(\mathbb{R}_+^*, d^*u)$, and throughout $\Re s > -1/2$,

$$\boxed{\mathcal{M}F(s) = \zeta(s+\tfrac{1}{2})\mathcal{M}f(s+\tfrac{1}{2}).} \tag{15.2}$$

Proof of (15.2). Partition the half-line into $((n-1)u, nu]$. Zero mass and bounded variation give

$$\left| u \sum_{n\geq 1} f(nu) \right| \leq u\,\mathrm{Var}(f),$$

which proves the L^2 bound at zero; compact support gives the upper cutoff. Absolute summation proves (15.2) for $\Re s > 1/2$. The pole at $s = 1/2$ is removed by zero mass, and analytic continuation gives the asserted half-plane. □

In particular, the Mellin transform of F vanishes at both coordinates attached to every nontrivial zero. Let P_I be restriction to $I_\lambda = [\lambda^{-1}, \lambda]$, and put

$$V_\lambda = P_I F, \qquad R_\lambda = (1 - P_I)F.$$

Theorem 15.1 (Moving-radical cancellation)**.** *For every semilocal core vector G supported in I_λ,*

$$\boxed{QW_\lambda(V_\lambda, G) = -QW(R_\lambda, G), \qquad QW_\lambda(V_\lambda, V_\lambda) = QW(R_\lambda, R_\lambda).} \tag{15.3}$$

The right sides are absolutely convergent zero-side values.

Proof. At every zero, the transform of R_λ is the negative of that of V_λ. Substitute this equality into the polarized zero-side formula, first with G, and then in both entries. □

If $H = V_\lambda^* * G$, Connes' semilocal trace formula [5] rewrites (15.3) without separating any source term:

$$\boxed{W_{0,2}(H) + \log(TW)H(1) + \mathrm{Tr}\Big(\vartheta(H)(1 - P_T^S - \widehat{P}_W^S)\Big) = -QW(R_\lambda, G).} \tag{15.4}$$

This is the exact joint prime–prime-power–Gamma–pole cancellation used below.

Poisson summation gives, for $0 < u < \lambda^{-1}$,

$$R_\lambda(u) = \mathcal{E}((1 - P_\lambda)\widehat{f})(u^{-1}) - \tfrac{1}{2}u^{1/2}f(0). \tag{15.5}$$

The standard midpoint convention is understood for zero-extended BV functions. Imposing also $f(0) = 0$ leaves only Fourier leakage.

Let $\psi_{j,\lambda}$, $j = 0, 4, 8, 12$, be normalized even prolate functions in the Fourier $+1$ sector, with compressed eigenvalues χ_j, and put $d_j = 1 - \chi_j^2$. The fixed-order asymptotics and compressed-Fourier identities are those used in [3]. The cofactor combination of the first three modes annihilates both rows

$$(\psi_j(0))_j, \qquad (\chi_j \psi_j(0))_j,$$

and therefore satisfies exactly $f(0) = \int f = 0$. On the two-dimensional four-mode constrained space, the angle form

$$\mathfrak{A}_\lambda(f, g) = \langle (1 - P_\lambda)\widehat{f}, (1 - P_\lambda)\widehat{g} \rangle$$

has generalized eigenvalues

$$\boxed{\alpha_{0,\lambda} \asymp d_4, \qquad \alpha_{1,\lambda} \asymp d_8, \qquad \frac{d_4}{d_8} \asymp \lambda^{-8},} \tag{15.6}$$

where

$$d_4 \asymp \lambda^9 e^{-4\pi\lambda^2}, \qquad d_8 \asymp \lambda^{17} e^{-4\pi\lambda^2}. \tag{15.7}$$

To verify (15.6), put $q_r = 1 - \chi_{4r}$. The constraints in the coordinates $x_r = \psi_{4r}(0)c_r$ are $\sum x_r = \sum q_r x_r = 0$, while $q_r/q_{r+1} = O(\lambda^{-8})$. The min–max test vectors

$$(q_1 - q_2, q_2 - q_0, q_0 - q_1, 0), \quad (0, q_2 - q_3, q_3 - q_1, q_1 - q_2)$$

give the matching upper bounds q_1, q_2; the two constraints give the corresponding lower bounds. Since $d_j \asymp q_j$, (15.6) follows.

The polynomial separation in (15.6) is more than sufficient for the Paley–Wiener weight in (13.3). The unresolved issue is transferring this angle hierarchy to the signed Weil operator and selecting the correct branch.

16. Why abstract geometry does not select that branch

Three natural selectors fail for exact reasons.

First, the completed semigroup is not positivity preserving. In the additive coordinate, for disjoint real tests f, g, the off-diagonal pole and Gamma kernels are respectively

$$2\cosh\frac{x-y}{2}, \qquad -c_\infty(|x-y|), \quad c_\infty(u) = \frac{e^{-u/2}}{1-e^{-2u}}.$$

The prime block is supported only when $|x-y| = \log p^k$. Choose two nonnegative disjoint bumps whose separation is near 1, avoiding every prime-power logarithm. Since

$$2\cosh(u/2) - c_\infty(u) > 0$$

there, their cross form is positive. This violates the first Beurling–Deny criterion $QW_L(f,g) \le 0$ for positivity preservation. The construction can be symmetrized, so it also rules out the natural pointwise cone in the even sector.

Second, adjoining finitely many derivatives of the full kernel does not repair the branch. Every transform in

$$\operatorname{span}\{K, K', \ldots, K^{(m)}\}$$

contains the common factor Ξ. A hypothetical off-line evaluation mode is therefore asymptotically orthogonal to every such fixed family. The same observation applies to every finite subspace of the complete co-Poisson radical.

Nor does real-rootedness prevent escape. If g_δ is the normalized indicator of $[-\delta, \delta]$, then

$$\psi_a(x) = 2^{-1/2}\big(g_\delta(x-a) + g_\delta(x+a)\big)$$

is positive, even and normalized, and $\widehat{\psi}_a(z) = \sqrt{2}\cos(az)\widehat{g}_\delta(z)$ has only real zeros. Nevertheless $\psi_a \rightharpoonup 0$ and becomes orthogonal to every fixed radical model as $a \to \infty$.

The exact many-vector obstruction is as follows.

Theorem 16.1 (Frame–residual conservation). *Let A be lower-semibounded, self-adjoint and of compact resolvent, and let $P_- = \mathbf{1}_{(-\infty,0)}(A)$. For $q_1, \ldots, q_M \in \operatorname{Dom}(A)$, define $Qc = \sum c_j q_j$, $D = AQ$, and*

$$a_-(Q) = \inf_{\substack{v \in \operatorname{Ran} P_- \\ \|v\|=1}} \|Q^* v\|^2.$$

If $a_-(Q) > 0$, then

$$\operatorname{Spec}(A|_{\operatorname{Ran} P_-}) \subset \left[-\frac{\|D^* P_-\|}{\sqrt{a_-(Q)}}, 0\right), \tag{16.1}$$

and for an orthonormal negative eigenbasis (ϕ_ℓ)*,*

$$\boxed{\sum_{j=1}^{M}\|P_-Aq_j\|^2=\sum_{\epsilon_\ell<0}\epsilon_\ell^2\sum_{j=1}^{M}|\langle\phi_\ell,q_j\rangle|^2\geq a_-(Q)\sum_{\epsilon_\ell<0}\epsilon_\ell^2.} \tag{16.2}$$

Proof. The negative space reduces A. Apply the lower frame inequality to Av and use $Q^*Av = D^*v$ to obtain (16.1). Expanding each P_-Aq_j in the negative eigenbasis and applying Parseval proves (16.2). ☐

Thus adding modes multiplies the useful overlap and the projected residual by the same amount. A fixed negative branch forces their quotient not to vanish.

There is an exact full-radical version. If S is any finite-dimensional subspace of the polarized Weil radical and $QW(h,h)<0$, then

$$h_S=h-P_Sh\perp S,\qquad QW(h_S,h_S)=QW(h,h)<0.$$

Thus no finite radical frame removes a negative direction; truncation can see it only through the aggregate exterior defect.

There is also a dimension mismatch in the presently available coercivity. Write $A_L=G_L+B_L$, where

$$\langle G_Lf,f\rangle\geq c_0\int\log(2+|t|)|\widehat{f}(t)|^2\frac{dt}{2\pi}-C_0\|f\|^2$$

and $\|B_L\|\leq b_L$. If $T_L=\exp(2(C_0+b_L)/c_0)$, a negative unit vector has at least half of its Fourier mass in $[-T_L,T_L]$. The time–frequency trace bound then gives

$$\dim\operatorname{Ran}P_-\leq\frac{2LT_L}{\pi}. \tag{16.3}$$

The absolute prime–pole estimate $b_L=O(\lambda)$ yields only $O(Le^{C\lambda})$ possible negative modes, whereas the Slepian family has dimension $O(\lambda^2)$. A signed estimate at logarithmic effective scale is therefore necessary.

17. The signed prime–Gamma branch gate

The trace projection itself contains an unavoidable sign choice. Let P,Q be orthogonal projections, and let a principal vector satisfy $PQPe=\chi^2e$. With $s=(1-\chi^2)^{1/2}$, the principal plane has

$$P=\begin{pmatrix}1&0\\0&0\end{pmatrix},\qquad Q=\begin{pmatrix}\chi^2&\chi s\\\chi s&s^2\end{pmatrix}.$$

Therefore, for $\mathcal{J}=I-P-Q$,

$$\boxed{\mathcal{J}^2=\chi^2I,\qquad\operatorname{Spec}\mathcal{J}=\{-\chi,+\chi\}.} \tag{17.1}$$

When the leakage $d=1-\chi^2$ tends to zero, both branches remain of order one. Prolate concentration fixes the square in (17.1), not the sign of its square root.

Let $q_L^{\mathrm{mov}}=V_\lambda/\|V_\lambda\|$ be the normalized first moving vector and $e_L^{\mathrm{ext}}=R_\lambda/\|V_\lambda\|$. Put

$$r_L^{\mathrm{mov}}=\langle A_Lq_L^{\mathrm{mov}},q_L^{\mathrm{mov}}\rangle,\quad r_L^{\perp}=(1-|q_L^{\mathrm{mov}}\rangle\langle q_L^{\mathrm{mov}}|)A_Lq_L^{\mathrm{mov}}.$$

The moving-radical identity gives the exact first-variation formulas

$$\boxed{\|r_L^{\perp}\|=\sup_{\substack{g\perp q_L^{\mathrm{mov}}\\\|g\|=1}}|QW(e_L^{\mathrm{ext}},g)|,\qquad r_L^{\mathrm{mov}}=QW(e_L^{\mathrm{ext}},e_L^{\mathrm{ext}}).} \tag{17.2}$$

The residual is linear in the exterior error, while the Rayleigh value is quadratic.

Even if one adds a polynomial logarithmic graph estimate, termwise control gives at best

$$\|r_L^{\perp}\| = O(\mathrm{poly}(\lambda)\sqrt{d_4}).$$

The branch scale required below is $o(\lambda^{-\sigma} d_8)$, $\sigma < 1/2$, and their ratio contains

$$\frac{\sqrt{d_4}}{\lambda^{-\sigma} d_8} \asymp \lambda^{\sigma - 25/2} e^{2\pi\lambda^2}. \tag{17.3}$$

Thus no polynomial regularity estimate reaches the selector: the complete linear variation must cancel.

Define the complementary bottom

$$\gamma_L^{\perp} = \inf_{\substack{g \perp q_L^{\mathrm{mov}} \\ \|g\|=1}} QW_L(g,g).$$

If $r_L^{\mathrm{mov}} < \gamma_L^{\perp}$, scalar Schur complementation gives

$$0 \le r_L^{\mathrm{mov}} - \epsilon_{0,L} \le \frac{\|r_L^{\perp}\|^2}{\gamma_L^{\perp} - r_L^{\mathrm{mov}}}, \qquad \epsilon_{1,L} \ge \gamma_L^{\perp}. \tag{17.4}$$

Open problem 17.1 (Gate SPG)**.** Prove, from the actual ordinary-prime source, that

$$r_L^{\mathrm{mov}} = O(d_4), \qquad \gamma_L^{\perp} \ge c d_8, \qquad \lambda^{\sigma} \frac{\|r_L^{\perp}\|}{\gamma_L^{\perp} - r_L^{\mathrm{mov}}} \longrightarrow 0 \quad (\sigma < 1/2). \tag{17.5}$$

Equivalently, for $\mathcal{H} = (q_L^{\mathrm{mov}})^* * g$, keep all terms coupled and prove

$$\sup_{\substack{g \perp q_L^{\mathrm{mov}} \\ \|g\|=1}} \left| W_{0,2}(\mathcal{H}) + \log(TW)\mathcal{H}(1) + \mathrm{Tr}\Big(\vartheta(\mathcal{H})(1 - P_T^S - \widehat{P}_W^S)\Big) \right| = o(\lambda^{-\sigma} d_8), \tag{17.6}$$

and the corresponding diagonal lower bound cd_8 on the orthogonal complement.

Gate SPG and Theorem 13.2 imply RH through Theorem 13.1, without identifying the entire ground transform. To deduce the stronger curvature limit (11.8), one additionally needs the distinct moving/model identification

$$\partial_z^2 \log \frac{\widehat{q}_L^{\mathrm{mov}}}{\widehat{k}_L} \longrightarrow 0. \tag{MI}$$

Neither implication is built into the other.

18. A local logarithmic theorem and the lost center

The logarithmic effective scale in Section 16 can be reached locally. Put

$$X_L = \lambda^2 = e^L, \qquad \mathcal{P}_{X_L}(t) = \sum_{n \le X_L} \frac{\Lambda(n)}{\sqrt{n}} n^{-it}.$$

Theorem 18.1 (Translated prime mean square)**.** *Uniformly for $U \in \mathbb{R}$ and $Y \ge 1$,*

$$\boxed{\int_{U-Y}^{U+Y} |\mathcal{P}_{X_L}(t)|^2 dt \le C(YL^2 + X_L L).} \tag{18.1}$$

In particular, for $Y = X_L/L$, the root-mean-square size is $O(L)$.

Proof. The translated Montgomery–Vaughan generalized Hilbert inequality [6] gives, for $a_n = \Lambda(n)n^{-1/2-iU}$,

$$\int_{-Y}^{Y} \left| \sum_{n \le X_L} a_n n^{-it} \right|^2 dt \le C \left(Y \sum |a_n|^2 + \sum n|a_n|^2 \right).$$

Chebyshev bounds and partial summation give

$$\sum_{n \le X_L} \frac{\Lambda(n)^2}{n} = O(L^2), \qquad \sum_{n \le X_L} \Lambda(n)^2 = O(X_L L),$$

which proves (18.1). □

If $v_1, \dots, v_d$ are orthonormal and supported in the additive interval of length L, Bessel's inequality gives $\sum_j |\widehat{v}_j(t)|^2 \le L$. On a block of length X_L/L, Cauchy–Schwarz and (18.1) therefore yield

$$\sum_j \int_B (2\Re \mathcal{P}_{X_L})_+ |\widehat{v}_j|^2 \frac{dt}{2\pi} \le CLX_L. \tag{18.2}$$

Near $|t| \asymp X_L/L$, the Gamma multiplier is at least cL. Hence an orthonormal family with a fixed fraction of its mass in that block and negative local prime–Gamma form has size $O(X_L) = O(\lambda^2)$. This is exactly the local Slepian budget.

The estimate does not globalize. It is uniform in U, and the absolute prime norm confines possible negative wells only to $|t| \le e^{C\lambda}$. Summing the local budget over all block centers returns $O(Le^{C\lambda})$. Nor can (18.1) exclude one negative well.

The loss is certified by twisting the coefficients:

$$a_n^{(\tau)} = \frac{\Lambda(n)}{\sqrt{n}} n^{i\tau}, \qquad \mathcal{P}_{X_L}^{(\tau)}(t) = \mathcal{P}_{X_L}(t-\tau).$$

All mean-square estimates remain unchanged, while the coherent peak $\sum_{n \le X_L} \Lambda(n)/\sqrt{n} \asymp 2\lambda$ moves to the arbitrary center τ. Thus magnitude estimates cannot select the fundamental branch. Gate SPG must use the untwisted locations and signs of the actual von Mangoldt phases jointly with Gamma and the polar block.

19. The centered prime–Gamma square

The center discarded by Theorem 18.1 can be restored exactly. Set

$$I_L = [-L/2, L/2], \qquad N = e^L = \lambda^2, \qquad w_n = \frac{\Lambda(n)}{\sqrt{n}}, \qquad a_n = \log n.$$

Extend $F \in L^2(I_L)$ by zero to $\mathbb{R}$, and let $(\tau_a F)(x) = F(x-a)$. This is full-line translation after zero extension, not periodic translation on I_L.

With

$$\theta(t) = \operatorname{Im} \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi,$$

the CCM normalization gives

$$Q_L(F,F) = \int_{\mathbb{R}} |\widehat{F}(t)|^2 \left(2\theta'(t) - 2\sum_{n\le N} w_n \cos(ta_n)\right) \frac{dt}{2\pi} + 2\,\mathrm{Re}\Big(\widehat{F}(i/2)\overline{\widehat{F}(-i/2)}\Big)\,. \tag{19.1}$$

Define the positive form

$$\mathcal{D}_N(F,G) := \sum_{n\le N} w_n \langle F-\tau_{a_n}F, G-\tau_{a_n}G\rangle + \sum_{k=0}^{\infty}\int_0^{\infty} e^{-2(k+1/4)x}\langle F-\tau_x F, G-\tau_x G\rangle\,dx \tag{19.2}$$

and the explicit scalar

$$\kappa_N = 2\sum_{n\le N}\frac{\Lambda(n)}{\sqrt{n}} - 2\theta'(0). \tag{19.3}$$

Theorem 19.1 (Exact centered square)**.** *On the CCM logarithmic Gamma-form domain,*

$$\boxed{Q_L(F,F) = \mathcal{D}_N(F,F) - \kappa_N\|F\|_2^2 + 2\,\mathrm{Re}\Big(\widehat{F}(i/2)\overline{\widehat{F}(-i/2)}\Big)\,.} \tag{19.4}$$

Moreover,

$$\boxed{\kappa_N = 2\sum_{n\le N}\frac{\Lambda(n)}{\sqrt{n}} + \gamma + \frac{\pi}{2} + 3\log 2 + \log\pi = 4\lambda + o(\lambda).} \tag{19.5}$$

Proof. Full-line Plancherel gives

$$\|F-\tau_a F\|_2^2 = 2\int_{\mathbb{R}} (1-\cos(at))|\widehat{F}(t)|^2\frac{dt}{2\pi}. \tag{19.6}$$

Writing $b_k = k+1/4$, the digamma expansion gives

$$\begin{aligned} 2\big(\theta'(t)-\theta'(0)\big) &= \sum_{k=0}^{\infty}\frac{(t/2)^2}{b_k\big(b_k^2+(t/2)^2\big)} \\ &= \sum_{k=0}^{\infty} 2\int_0^{\infty} e^{-2b_k x}(1-\cos(tx))\,dx. \end{aligned} \tag{19.7}$$

Hence the Fourier multiplier of $\mathcal{D}_N$ is

$$2\sum_{n\le N} w_n(1-\cos(ta_n)) + 2(\theta'(t)-\theta'(0)).$$

Subtracting (19.3) gives the multiplier in (19.1), which proves (19.4). Finally,

$$2\theta'(0) = \psi(1/4) - \log\pi = -\gamma - \frac{\pi}{2} - 3\log 2 - \log\pi,$$

and PNT with partial summation gives $\sum_{n\le N}\Lambda(n)/\sqrt{n} = 2\sqrt{N} + o(\sqrt{N})$. $\square$

Corollary 19.2 (Unique fundamental center)**.** *For*

$$m_N(t) = 2\theta'(t) - 2\sum_{n\le N} w_n \cos(t\log n),$$

one has

$$\boxed{\begin{aligned} m_N(t) - m_N(0) &= 2\sum_{n\le N} w_n(1-\cos(t\log n)) \\ &\quad + 2(\theta'(t)-\theta'(0)) \ge 0, \end{aligned}} \tag{19.8}$$

with strict inequality for $t \neq 0$.

Proof. Every summand is nonnegative, and the Gamma series in (19.7) is strictly positive away from zero. ☐

Thus the artificial coefficient twist in Section 18 is excluded by the exact source identity: it moves the prime well without moving the Gamma minimum.

There is also a pure-gauge representation. Put

$$\phi_N(t) = \theta(t) - \sum_{n\le N} \frac{\Lambda(n)}{\sqrt{n}\log n}\sin(t\log n), \quad U_N = e^{i\phi_N}, \quad \mathsf{X} = -i\partial_t.$$

Then, on the smooth core,

$$2(U_N^* \mathsf{X} U_N - \mathsf{X}) = M_{m_N}. \tag{19.9}$$

This identity does not remove the finite-interval boundary or the polar channel.

In the even sector,

$$\widehat{F}(i/2) = \widehat{F}(-i/2) = \langle h_L, F\rangle, \qquad h_L = \mathbf{1}_{I_L}\cosh(x/2),$$

and therefore

$$\boxed{Q_L^+(F,G) = \mathcal{D}_N(F,G) - \kappa_N\langle F,G\rangle + 2\overline{\langle h_L,F\rangle}\langle h_L,G\rangle.} \tag{19.10}$$

If $\mathcal{L}_N^+$ is associated with

$$\ell_N^+(F,G) = \mathcal{D}_N(F,G) + 2\overline{\langle h_L,F\rangle}\langle h_L,G\rangle,$$

then

$$\boxed{A_L^+ = \mathcal{L}_N^+ - \kappa_N I, \qquad \mathcal{L}_N^+ \ge 0.} \tag{19.11}$$

Consequently Gate SPG can be restated as a quantitative rank-one spectral theorem for the explicit positive operator $\mathcal{L}_N^+$. For the moving vector $q_L = q_L^{\mathrm{mov}}$, its cross part is

$$\sup_{\substack{g\perp q_L\\ \|g\|=1}} \left|\mathcal{D}_N(q_L,g) + 2\overline{\langle h_L,q_L\rangle}\langle h_L,g\rangle\right| = o(\lambda^{-\sigma}d_8), \quad \sigma < \frac{1}{2}, \tag{19.12}$$

and its complementary part is

$$\inf_{\substack{g\perp q_L\\ \|g\|=1}} \left(\mathcal{D}_N(g,g) + 2|\langle h_L,g\rangle|^2\right) \ge \kappa_N + cd_8. \tag{19.13}$$

If the separate Gate-SPG Rayleigh estimate $r_L^{\mathrm{mov}} = O(d_4)$ is proved, then $\ell_N^+(q_L,q_L) = \kappa_N + O(d_4)$. Together with that estimate, (19.12)–(19.13) assert that q_L is a first-level $\mathcal{L}_N^+$-quasimode to the required form-dual accuracy and that the next level clears the explicit threshold. Positivity of $\mathcal{L}_N^+$ alone does not imply these assertions.

For qualitative nonnegativity there is a narrower conditional rank-one criterion. Assume the force-bearing inertia statement

$$\lambda_1(\mathcal{D}_N) < \kappa_N < \lambda_2(\mathcal{D}_N).$$

Then rank-one inertia gives

$$Q_L^+ \geq 0 \quad \Longleftrightarrow \quad 1 + 2\langle h_L, (\mathcal{D}_N - \kappa_N)^{-1} h_L \rangle \leq 0. \tag{19.14}$$

At finite Galerkin level the second factor is the determinant ratio

$$\frac{\det(\mathcal{L}_N^+ - z)}{\det(\mathcal{D}_N - z)} = 1 + 2\langle h_L, (\mathcal{D}_N - z)^{-1} h_L \rangle. \tag{19.15}$$

The sign in (19.14), rather than existence of the square in (19.4), is arithmetic spectral information. Criterion (19.14) concerns only qualitative nonnegativity under the additional inertia hypothesis; it does not imply the quantitative residual and d_8 gap in (19.12)–(19.13).

Remark 19.3 (Form domain and boundary). The Gamma jump density in (19.2) behaves as $dx/(2x)$ near zero. Define the logarithmic form domain as the completion of $C_c^\infty(I_L)$ under

$$\|F\|_2^2 + \int_{\mathbb{R}} \log(2 + |t|) |\widehat{F}(t)|^2 \, dt.$$

Then the identity extends to that domain by closure. For translations comparable with L, zero extension creates a boundary loss which cancels the $O(\lambda)$ constant in (19.3). Replacing τ_a by periodic translation gives a false formula.

20. Exact PNT-discrepancy compensation

The same cancellation has a complementary Stieltjes form. Let f, g be smooth, zero-extended functions on I_L, and put

$$H(u) = \langle f, \tau_u g \rangle, \qquad F(u) = H(u) + H(-u), \qquad \Psi(u) = \psi(e^u). \tag{20.1}$$

In this section $Q_L = QW_L$ denotes the same semilocal CCM form. For a Gate-SPG cross term $g \perp f$, one has $F(0) = 0$.

The polar, Euler and off-diagonal Gamma terms are respectively

$$\begin{aligned} P(f,g) &= \int_0^L F(u)(e^{u/2} + e^{-u/2}) \, du, \\ E(f,g) &= -\int_{(0,L]} F(u) e^{-u/2} \, d\Psi(u), \\ G(f,g) &= -\int_0^L F(u) \frac{e^{-u/2}}{1 - e^{-2u}} \, du. \end{aligned} \tag{20.2}$$

Theorem 20.1 (Complete source compensation)**.** *For every smooth-core Gate-SPG cross term,*

$$\boxed{\begin{aligned} QW_L(f,g) = &-\int_0^L F(u) \frac{e^{-5u/2}}{1 - e^{-2u}} \, du \\ &- \int_{(0,L]} F(u) e^{-u/2} \, d\big(\psi(e^u) - e^u\big). \end{aligned}} \tag{20.3}$$

Proof. Since $d(e^u) = e^u\,du$,

$$P + E = \int_0^L F(u)e^{-u/2}\,du - \int_{(0,L]} F(u)e^{-u/2}\,d(\Psi(u) - e^u). \tag{20.4}$$

The first term in (20.4) combines with Gamma through

$$e^{-u/2} - \frac{e^{-u/2}}{1-e^{-2u}} = -\frac{e^{-5u/2}}{1-e^{-2u}}.$$

This proves (20.3). The kernel is $1/(2u) - 3/4 + O(u)$ near zero, and $F(0) = 0$ makes the integral finite. □

The continuous PNT main measure has therefore cancelled the growing polar branch exactly, while the remaining polar branch has combined with Gamma. The sole unknown arithmetic distribution in (20.3) is $d(\psi(e^u) - e^u)$. The explicit locally singular integral also has no sign for a general cross correlation and must be controlled jointly. Extension of the two separated integrals from the smooth core to arbitrary Gate-SPG form-domain pairs requires a separate continuity argument.

The full quadratic completion. For a zero-extended f put

$$H_f(u) = \langle f, \tau_u f\rangle, \qquad F_f(u) = H_f(u) + H_f(-u),$$

and define

$$\begin{aligned} \mathcal{E}_*(f) &:= \int_0^\infty \|f - \tau_u f\|_2^2 \frac{e^{-5u/2}}{1-e^{-2u}}\,du, \\ \mathcal{A}_\Delta(f) &:= \int_{(0,L]} F_f(u)e^{-u/2}\,d\big(\psi(e^u) - e^u\big), \\ c_* &:= \gamma + \frac{\pi}{2} + 3\log 2 + \log\pi - 4 > 0. \end{aligned} \tag{20.5}$$

Theorem 20.2 (Exact quadratic prime–Gamma–pole compensation)**.** *On the smooth form core,*

$$\boxed{QW_L(f,f) = \mathcal{E}_*(f) - c_*\|f\|_2^2 - \mathcal{A}_\Delta(f).} \tag{20.6}$$

Proof. Let

$$d\nu_p(u) = e^{-u/2}d\psi(e^u), \quad d\nu_0(u) = e^{u/2}du, \quad d\nu_e(u) = e^{-u/2}du.$$

Expanding the prime jump form in Theorem 19.1 and subtracting its scalar part gives $-\int F_f\,d\nu_p + 2\theta'(0)\|f\|^2$. The growing polar branch cancels ν_0, leaving

$$\int_0^L F_f\,d\nu_e - \mathcal{A}_\Delta(f).$$

Since $\nu_e(0,\infty) = 2$,

$$\int_0^L F_f\,d\nu_e = 4\|f\|_2^2 - \int_0^\infty \|f - \tau_u f\|_2^2\,d\nu_e(u).$$

Finally,

$$\frac{e^{-u/2}}{1-e^{-2u}} - e^{-u/2} = \frac{e^{-5u/2}}{1-e^{-2u}},$$

and $4 + 2\theta'(0) = -c_*$. Combining the identities proves (20.6). □

Thus qualitative positivity of the completed form is exactly the literal-prime inequality

$$\boxed{\mathcal{A}_\Delta(f) \le \mathcal{E}_*(f) - c_*\|f\|_2^2 \quad \text{for every test } f.} \tag{20.7}$$

Equation (20.7) is not proved here. It is the one-sided signed theorem left after the continuous PNT main term, Gamma and both polar branches have been cancelled algebraically.

This reduction is already nontrivial on a normalized Fejer packet

$$f_{L,U}(x) = L^{-1/2} e^{iUx}\mathbf{1}_{I_L}(x).$$

This is a diagonal calculation, with $F(0) = 2$, and is therefore not an application of the cross identity (20.3). Define

$$\begin{aligned} S_L(U) &= \sum_{n \le e^L} \frac{\Lambda(n)}{\sqrt{n}} \left(1 - \frac{\log n}{L}\right) \cos(U \log n), \\ M_\pm(L,U) &= \int_0^L e^{\pm u/2} \left(1 - \frac{u}{L}\right) \cos(Uu)\, du, \\ \Delta_L(U) &= S_L(U) - M_+(L,U). \end{aligned} \tag{20.8}$$

Put

$$A_+(L,U) = \frac{2\sinh((1/2+iU)L/2)}{\sqrt{L}\,(1/2+iU)}.$$

Direct integration gives

$$P_L(U) = 2\,\mathrm{Re}\, A_+(L,U)^2 = 2(M_+ + M_-), \tag{20.9}$$

or, equivalently,

$$\begin{aligned} P_L(U) = \frac{4}{L(U^2+1/4)^2} \Big[&(1/4 - U^2)(\cosh(L/2)\cos(UL) - 1) \\ &+ U \sinh(L/2)\sin(UL) \Big]. \end{aligned} \tag{20.9a}$$

This polar value is not sign-definite. Define the full archimedean packet value by

$$A_{\infty,L}(U) = \int_{\mathbb{R}} 2\theta'(t) |\widehat{f}_{L,U}(t)|^2 \, \frac{dt}{2\pi}. \tag{20.9b}$$

Then

$$\boxed{Q_L(f_{L,U}, f_{L,U}) = A_{\infty,L}(U) + 2M_-(L,U) - 2\Delta_L(U).} \tag{20.10}$$

The scalar signed target is therefore

$$\Delta_L(U) \le M_-(L,U) + \tfrac{1}{2} A_{\infty,L}(U). \tag{20.11}$$

It is not proved here.

There is, however, an unconditional far-center result. Put

$$c_\Gamma = \frac{1}{4\pi}, \quad C_\Gamma = \sup_t \{c_\Gamma \log(2+|t|) - 2\theta'(t)\}, \quad C_\psi = \sup_{x \ge 1} \frac{\psi(x)}{x},$$

and

$$m_0 = \frac{2}{\pi} \int_0^1 \left(\frac{\sin y}{y}\right)^2 dy \ge \frac{2\sin^2(1)}{\pi}.$$

The Fejer kernel of $f_{L,U}$ has mass m_0 on $|t - U| \le 2/L$, while Stieltjes integration by parts gives

$$|S_L(U)| \le S_L(0) \le \frac{4C_\psi e^{L/2}}{L}.$$

The exact polar formula (20.9a) also gives

$$\begin{aligned} P_L(U) &\geq -\frac{4}{L}\left[\frac{\cosh(L/2)+1}{U^2+1/4}+\frac{|U|\sinh(L/2)}{(U^2+1/4)^2}\right] \\ &\geq -\frac{5e^{L/2}}{LU^2} \qquad (L\geq 2,\ |U|\geq 4). \end{aligned}$$

It follows that, for $L \geq 2$ and $|U| \geq 4$,

$$\boxed{Q_L(f_{L,U}, f_{L,U}) \geq m_0 c_\Gamma \log(2+|U|/2) - C_\Gamma - \frac{8C_\psi e^{L/2}}{L} - \frac{5e^{L/2}}{LU^2}.} \tag{20.12}$$

Thus these packets are nonnegative once

$$\log(2+|U|/2) \geq \frac{C_\Gamma + (8C_\psi + 5/16)\lambda/L}{m_0 c_\Gamma}. \tag{20.13}$$

This improves the crude possible-center range from $\exp(C\lambda)$ to $\exp(C\lambda/L)$ for this packet family. It still leaves every fixed center as $L \to \infty$.

Finally, exact polar tapering does not create second-moment center decay. For

$$\mathcal{P}_L^{\triangle}(t) = \sum_{n\leq e^L} \frac{\Lambda(n)}{\sqrt{n}}\left(1-\frac{\log n}{L}\right) n^{-it},$$

The classical zero-free-region PNT and the two-sided Montgomery–Vaughan theorem give, uniformly in U,

$$\frac{1}{2Y}\int_{U-Y}^{U+Y} |\mathcal{P}_L^{\triangle}(t)|^2 dt = \frac{L^2}{12} + O(1), \qquad Y = \frac{e^L}{L}. \tag{20.14}$$

The remaining theorem must therefore be genuinely one-sided and signed, not another RMS estimate.

21. The Picone gate and the positive-jump obstruction

The positive square in Section 19 suggests a nonlocal ground-state transform. It exists exactly, but does not supply the missing constant.

Write

$$\nu_N = \sum_{n\leq N} \frac{\Lambda(n)}{\sqrt{n}} \delta_{\log n} + \frac{e^{-u/2}}{1-e^{-2u}}\, du. \tag{21.1}$$

Let $v > 0$ on I_L, extend v by zero, and write $r = f/v$ on I_L.

Theorem 21.1 (Zero-extension Picone identity)**.** *On a common smooth form core,*

$$\boxed{\begin{aligned} \mathcal{D}_N(f,f) = &\int_{(0,\infty)}\int_{\mathbb{R}} v(x)v(x-u)|r(x)-r(x-u)|^2\, dx\, \nu_N(du) \\ &+ \int_{I_L} \frac{(\mathcal{D}_N v)(x)}{v(x)} |f(x)|^2\, dx. \end{aligned}} \tag{21.2}$$

Proof. For each shift use

$$|v_x r_x - v_y r_y|^2 = v_x v_y |r_x - r_y|^2 + (v_x - v_y)(v_x|r_x|^2 - v_y|r_y|^2).$$

Set $y = x - u$, integrate, and change variables in the second half of the last term. It becomes

$$\int_{I_L} \frac{2v(x) - v(x-u) - v(x+u)}{v(x)} |f(x)|^2 \, dx.$$

Integration against ν_N proves (21.2), including the zero-extension boundary killing. $\square$

If $\varphi_1 > 0$ is the normalized ground state of $\mathcal{D}_N$, with eigenvalue μ_1, then

$$\mathcal{D}_N(f,f) - \mu_1 \|f\|_2^2 = \iint \varphi_1(x)\varphi_1(x-u) \left| \frac{f(x)}{\varphi_1(x)} - \frac{f(x-u)}{\varphi_1(x-u)} \right|^2 dx \, \nu_N(du). \tag{21.3}$$

Consequently $\lambda_2(\mathcal{D}_N) \geq \kappa_N$ is exactly the weighted Poincare inequality

$$\iint \varphi_1(x)\varphi_1(x-u)|r(x) - r(x-u)|^2 dx \, \nu_N(du) \geq (\kappa_N - \mu_1) \int \varphi_1^2 |r|^2, \tag{21.4}$$

for $\int \varphi_1^2 r = 0$. The transform is lossless: its required constant is the missing second-level bound itself.

The resolvent sign is equally exact. Put $B_N = \mathcal{D}_N - \kappa_N I$, assume the strict one-negative-level inertia from (19.14), and set $r_N = \langle h_L, B_N^{-1} h_L \rangle < 0$. Then

$$\inf_{\langle h_L, f \rangle = 1} B_N(f,f) = \frac{1}{r_N}, \tag{21.5}$$

so

$$1 + 2r_N \leq 0 \quad \Longleftrightarrow \quad \inf_{\langle h_L, f \rangle = 1} B_N(f,f) \geq -2. \tag{21.6}$$

Together with $B_N \geq 0$ on $h_L^\perp$, this is precisely the original completed even Weil inequality.

The following countermodel shows why generic jump positivity cannot prove the inertia premise.

Proposition 21.2 (Positive one-atom obstruction)**.** *Fix $a > 0$ and $M \geq 6$. There is a positive jump form with the same Gamma channel, zero-extension boundary killing and polar rank-one perturbation whose scalar-shifted compression has at least $\lfloor M/2 \rfloor$ negative eigenvalues.*

Proof. Choose normalized smooth bumps $\phi_j = \tau_{(j-1)a}\phi_1$, $1 \leq j \leq M$, with disjoint supports of width below $a/3$, inside a sufficiently long interval. Add the positive atom $w\delta_a$ and subtract its scalar contribution $2wI$, as in (19.3). On their span the atomic part is

$$-wA_M,$$

where A_M is the path adjacency matrix, with eigenvalues

$$-2w \cos \frac{k\pi}{M+1}, \qquad 1 \leq k \leq M.$$

The fixed Gamma, scalar and polar compressions have norm independent of w. Weyl's inequality therefore leaves at least $\lfloor M/2 \rfloor$ negative levels for sufficiently large w. $\square$

Thus positivity of the jump measure, Picone algebra, Hardy completion and rank-one perturbation do not select the fundamental branch. The actual locations and magnitudes $\Lambda(n)/\sqrt{n}$ must enter through the literal compensated inequality (20.7), quantitatively strengthened to the moving-vector residual and d_8 scale of Gate SPG.

22. Smooth full-kernel operator quasimodes

The moving co-Poisson construction retains the finite-cutoff geometry of CCM. For the direct positivity problem there is a second route: truncate Riemann's full kernel only after using its exact radical property.

Let K be the even positive Schwartz function normalized by

$$\widehat{K}(z) = \Xi(z). \tag{22.1}$$

Write $a = L/2$, $\lambda = e^a$, and choose smooth even cutoffs χ_a equal to one on $[-a+1, a-1]$ and supported in $(-a, a)$, with all fixed derivative seminorms bounded uniformly in a. Put

$$v_L = \chi_a K, \qquad q_L^K = \frac{v_L}{\|v_L\|_2}. \tag{22.2}$$

The theta expansion and modular symmetry imply, for every fixed j,

$$|K^{(j)}(x)| \le C_j e^{M_j|x|} e^{-c_j e^{2|x|}}. \tag{22.3}$$

Consequently, if $e_L = v_L - K$, repeated integration by parts gives

$$\sup_{|\eta|\le 1/2} |\widehat{e}_L(t+i\eta)| \le C_r \lambda^{M_r} e^{-c_r\lambda^2}(1+|t|)^{-r}. \tag{22.4}$$

Theorem 22.1 (Full-kernel operator quasimode)**.** *For suitable positive constants C, M, c,*

$$\|A_L^+ v_L\|_2 \le C\sqrt{L\lambda}\,\lambda^M e^{-c\lambda^2}. \tag{22.5}$$

In particular,

$$\|A_L^+ q_L^K\|_2 \longrightarrow 0. \tag{22.6}$$

Proof. At every nontrivial zero z_ρ in the additive spectral coordinate, $\widehat{K}(z_\rho) = 0$. Thus the polarized Weil form satisfies

$$QW(K, g) = 0, \tag{22.7}$$

and $QW(v_L, g) = QW(e_L, g)$. For $g \in L^2([-a, a])$,

$$|\widehat{g}(z_\rho)| \le \sqrt{L\lambda}\,\|g\|_2, \tag{22.8}$$

because $|\operatorname{Im} z_\rho| < 1/2$. The local Riemann–von Mangoldt estimate gives $\sum_\rho (1+|\gamma|)^{-r} < \infty$ for $r > 1$. Combining (22.4), the polarized zero sum and (22.8) proves

$$|QW(v_L, g)| \le C\sqrt{L\lambda}\,\lambda^M e^{-c\lambda^2}\|g\|_2.$$

The Riesz representative of this functional is $A_L^+ v_L$. Finally $v_L \to K$ in L^2, so its norm stays bounded away from zero. □

Theorem 22.1 closes the residual half of the finite-cutoff branch criterion. It does not supply its complementary spectral floor. Ordinary cyclicity of translates and derivatives of K cannot supply that floor at the required scale. Indeed, if an off-line zero $\rho = \beta + i\gamma$, $\beta > 1/2$, exists and a fixed compactly supported h detects it, then every truncated radical approximant w_L with $\widehat{w}_L(z_\rho) = o(1)$ satisfies

$$\|h - w_L\|_2^2 \gg_{h,\rho} \lambda^{-2(\beta-1/2)}, \qquad \kappa_{e^L}\|h - w_L\|_2^2 \gg_{h,\rho} \lambda^{2(1-\beta)} \longrightarrow \infty, \tag{22.9}$$

where $\kappa_{e^L} = 4\lambda + o(\lambda)$. Thus qualitative L^2-density has exactly the wrong conditioning for the semilocal inertia problem.

23. A cutoff-free normalization of the classical Weil criterion

Weil positivity is a classical criterion equivalent to RH [1]. Many later works give other analytic, Hilbert-space and dynamical equivalences; in particular Sun derives an explicit integral inequality from $|\xi(\sigma-it)|^2$ and an equivalent infinite orthogonalization-time statement on Wiener space [2]. The purpose of this section is not to add another logical equivalent. It is to express the same classical Weil form in the exact ground coordinate needed to compare it with the finite CCM–Suzuki operators and with the literal prime-power decomposition of the preceding sections.

The full kernel removes the moving branch from the algebra before any estimate is made. Put

$$h(x) = \cosh(x/2), \qquad c_K = \int_{\mathbb{R}} h(x)K(x)\,dx = \widehat{K}(i/2) = \xi(0) = \frac{1}{2} \tag{23.1}$$

and define the probability measure

$$d\mu_K(x) = \frac{h(x)K(x)}{c_K}\,dx = 2\cosh(x/2)K(x)\,dx. \tag{23.2}$$

The complete positive jump measure is

$$\nu_\zeta(du) = \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}}\delta_{\log n}(du) + \frac{e^{-u/2}}{1-e^{-2u}}\,du, \qquad u>0. \tag{23.3}$$

For even $r \in C_c^\infty(\mathbb{R})$, set

$$\mathscr{E}_K(r) = \int_{(0,\infty)}\int_{\mathbb{R}} K(x)K(x-u)|r(x)-r(x-u)|^2\,dx\,\nu_\zeta(du). \tag{23.4}$$

Theorem 23.1 (Full-kernel Doob–variance identity)**.** *For every even complex $r \in C_c^\infty(\mathbb{R})$,*

$$\boxed{QW(Kr,Kr) = \mathscr{E}_K(r) - \frac{1}{2}\operatorname{Var}_{\mu_K}(r).} \tag{23.5}$$

Proof. Use a common smooth spatial regulator and the same prime cutoff in the two physical-side Weil forms $QW(Kr,Kr)$ and $QW(K,K|r|^2)$. At every retained jump, with $y=x-u$, the pointwise identity

$$|K_xr_x - K_yr_y|^2 - (K_x-K_y)(K_x|r_x|^2 - K_y|r_y|^2) = K_xK_y|r_x-r_y|^2 \tag{23.6}$$

gives (23.4). The scalar centering terms cancel. In the even sector the two polar evaluations contribute

$$2\left|\int hKr\right|^2 - 2\left(\int hK\right)\left(\int hK|r|^2\right) = -2c_K^2\operatorname{Var}_{\mu_K}(r). \tag{23.7}$$

Since $2c_K^2 = 1/2$, this is the second term of (23.5).

The zero-side radical identity $QW(K,g)=0$, applied to $g=K|r|^2$, removes the second regulated Weil form in the limit. The Gamma contribution is integrable at zero because the squared difference in (23.4) is $O(u^2)$, and the prime sum converges after the spatial regulator is removed because one translated K-factor has double-exponential decay. Monotone convergence applies to the nonnegative prime energy. Removing the two regulators therefore proves (23.5). □

The jump form (23.4) is closable on $L^2(\mu_K)$. Indeed, define the positive symmetric σ-finite measure J_K by

$$\int F(x,y)\,J_K(dx,dy) = \frac{1}{2}\int_{(0,\infty)}\int_{\mathbb{R}} K(x)K(x-u)\big(F(x,x-u)+F(x-u,x)\big)\,dx\,\nu_\zeta(du). \tag{23.7a}$$

Then (23.4) is the standard jump form $\int |r(x)-r(y)|^2 J_K(dx,dy)$, initially on its smooth core. Let $\mathcal{L}_K$ denote the associated nonnegative self-adjoint operator, whose weak action on the core is

$$(\mathcal{L}_K r)(x) = \frac{c_K}{h(x)} \int_{(0,\infty)} \big[K(x-u)(r(x)-r(x-u)) + K(x+u)(r(x)-r(x+u))\big]\,\nu_\zeta(du). \tag{23.7b}$$

Let and put

$$\mathcal{H}_{0,+} = \left\{ r \in L^2(\mu_K) : r(-x) = r(x),\ \int r\, d\mu_K = 0 \right\}. \tag{23.8}$$

Corollary 23.2 (Exact spectral floor)**.** *The following statements are equivalent:*

$$\text{RH}; \qquad \mathscr{E}_K(r) \geq \frac{1}{2}\operatorname{Var}_{\mu_K}(r) \quad (r \in C^\infty_{c,\text{even}}); \qquad \inf \sigma(\mathcal{L}_K|_{\mathcal{H}_{0,+}}) \geq \frac{1}{2}. \tag{23.9}$$

Proof. Every compactly supported smooth even test f has the unique form $f = Kr$, since $K > 0$. By the bilateral parity theorem, failure of RH already produces a negative even Weil test. Theorem 23.1 therefore converts the even Weil criterion exactly into the middle inequality of (23.9). Closing the form gives the spectral statement. □

Corollary 23.2 is a reformulation of the classical Weil criterion. It is not, by itself, progress toward proving the missing positivity, nor does it establish the strong-resolvent convergence proposed in the CCM–Suzuki program. Its role here is bookkeeping: it places the remaining sign question in the same full-kernel coordinate as the exact theta and von Mangoldt decompositions below.

The threshold is not an arbitrary estimate. It already contains an infinite radical family.

Proposition 23.3 (Infinite threshold family)**.** *For $m \geq 1$, define*

$$r_m = \frac{K^{(2m)}}{K} - 4^{-m}. \tag{23.10}$$

Then $r_m \in \mathcal{H}_{0,+}$ belongs to the extended form domain and is a weak eigenfunction of $\mathcal{L}_K$ with eigenvalue $1/2$. The family $(r_m)_{m\geq1}$ is linearly independent. Consequently $1/2$ is an infinite-multiplicity, hence essential, spectral point.

Proof. Integration by parts and $h^{(2m)} = 4^{-m}h$ give

$$\int \frac{K^{(2m)}}{K}\, d\mu_K = 4^{-m},$$

so r_m has mean zero. Moreover

$$\widehat{K^{(2m)}}(z) = (-1)^m z^{2m}\Xi(z),$$

and both $K^{(2m)}$ and K lie in the polarized Weil radical. Polarizing (23.5) therefore yields

$$\mathscr{E}_K(r_m, s) = \frac{1}{2}\langle r_m, s\rangle_{L^2(\mu_K)} \tag{23.11}$$

for every even test s, and cutoff approximation extends the identity to the form domain. Finally, the theta asymptotics make $K^{(2m)}/K$ a function with a distinct highest growth degree in $e^{2|x|}$ for each m; hence no nontrivial finite linear combination can vanish. □

Proposition 23.3 supplies an important qualification. The currently proved equivalence is the absence of *any* spectrum below 1/2. Calling every possible subthreshold point a bound state requires one additional structural theorem,

$$\inf \sigma_{\mathrm{ess}}(\mathcal{L}_K|_{\mathcal{H}_{0,+}}) \geq \frac{1}{2}, \tag{23.12}$$

after which the remaining exclusion would indeed concern only discrete bound states.

24. Atomwise rigidity and the all-source sharp gate

For $u > 0$, write

$$\mathcal{J}_u(r) = \int_{\mathbb{R}} K(x)K(x-u)|r(x) - r(x-u)|^2 \, dx. \tag{24.1}$$

Then

$$\mathscr{E}_K(r) = \mathscr{E}_\Gamma(r) + \sum_{m\geq 2} \frac{\Lambda(m)}{\sqrt{m}} \mathcal{J}_{\log m}(r). \tag{24.2}$$

This is an increasing hierarchy in the set of retained prime-power atoms. Positivity of the omitted tail, however, cannot be used to reduce the problem to finitely many atoms.

Theorem 24.1 (Every von Mangoldt atom is load-bearing)**.** *Let S omit at least one prime power, and define*

$$\mathscr{E}_S(r) = \mathscr{E}_\Gamma(r) + \sum_{m\in S} \frac{\Lambda(m)}{\sqrt{m}} \mathcal{J}_{\log m}(r). \tag{24.3}$$

Then the sharp inequality $\mathscr{E}_S(r) \geq \frac{1}{2}\operatorname{Var}_{\mu_K}(r)$ is false. More precisely, there are even compactly supported smooth r_a for which the reverse strict inequality holds for every sufficiently large a.

Proof. Take

$$f_a = \chi_a K'', \qquad r_a = \frac{f_a}{K}. \tag{24.4}$$

The transform $\widehat{K''}(z) = -z^2\Xi(z)$ and the exterior estimate of Theorem 22.1 give $QW(f_a, f_a) \to 0$. For every fixed $u > 0$,

$$\mathcal{J}_u(r_a) \longrightarrow \mathcal{J}_u(K''/K) > 0. \tag{24.5}$$

Strict positivity follows because equality would make K''/K u-periodic, whereas its theta asymptotic is unbounded. Subtracting the omitted atoms from (23.5), and retaining one omitted m_0, gives

$$\limsup_{a\to\infty} \left(\mathscr{E}_S(r_a) - \frac{1}{2}\operatorname{Var}_{\mu_K}(r_a)\right) \leq -\frac{\Lambda(m_0)}{\sqrt{m_0}} \mathcal{J}_{\log m_0}(K''/K) < 0. \tag{24.6}$$

□

The theta series contains an exact dilation which exposes the missing coupling. Suppressing the common positive normalization, put

$$k_j(x) = \pi j^2 e^{5x/2}\left(2\pi j^2 e^{2x} - 3\right) e^{-\pi j^2 e^{2x}}, \qquad K(x) = \sum_{j\geq 1} k_j(x) \quad (x \geq 0). \tag{24.7}$$

For every integer $n \geq 2$,

$$\boxed{k_{nj}(x - \log n) = n^{-1/2} k_j(x).} \tag{24.8}$$

Consequently, when $x \geq \log n$,

$$K(x - \log n) = n^{-1/2} K(x) + \sum_{\substack{j \geq 1 \\ n \nmid j}} k_j(x - \log n). \tag{24.9}$$

Dropping the second term and the central crossing region $0 < x < \log n$ produces a valid lower bound, but Theorem 24.1 shows that it is strictly too small at the sharp constant. Thus the all-prime theorem must preserve simultaneously:

(1) every atom $\Lambda(p^k)$;
(2) the theta indices divisible by the corresponding dilation;
(3) the nondivisible remainder in (24.9);
(4) the central crossing region;
(5) the Gamma jump continuum and the polar variance.

The arithmetic identity

$$\sum_{d \mid j} \Lambda(d) = \log j \tag{24.10}$$

shows the natural point at which the divisible theta channel can couple to the logarithmic derivative implicit in the Gamma term. It does not control the nondivisible and crossing channels separately; those channels cannot be discarded, by (24.6).

There is, however, an exact reorganization which retains all of those channels. Write $\mathscr{E}_p = \mathscr{E}_K - \mathscr{E}_\Gamma$.

Proposition 24.2 (Exact half-line theta–divisor lift)**.** *For every even r in the test core,*

$$\mathscr{E}_p(r) = \mathscr{E}_p^{\mathrm{same}}(r) + \mathscr{E}_p^{\mathrm{cross}}(r), \tag{24.11}$$

where

$$\begin{aligned} \mathscr{E}_p^{\mathrm{same}}(r) &= 2 \int_0^\infty K(x) \sum_{\ell \geq 2} k_\ell(x) \sum_{\substack{d \mid \ell \\ d \geq 2}} \Lambda(d) |r(x) - r(x + \log d)|^2 \, dx, \\ \mathscr{E}_p^{\mathrm{cross}}(r) &= \sum_{d \geq 2} \frac{\Lambda(d)}{\sqrt{d}} \int_0^{\log d} K(t) K(\log d - t) |r(t) - r(\log d - t)|^2 \, dt. \end{aligned} \tag{24.12}$$

The Gamma energy has the parallel decomposition

$$\begin{aligned} \mathscr{E}_\Gamma^{\mathrm{same}}(r) &= 2 \int_0^\infty \int_0^\infty K(x) K(x+u) |r(x) - r(x+u)|^2 \, dx \, \nu_\Gamma(du), \\ \mathscr{E}_\Gamma^{\mathrm{cross}}(r) &= \int_0^\infty \int_0^u K(t) K(u-t) |r(t) - r(u-t)|^2 \, dt \, \nu_\Gamma(du), \end{aligned} \tag{24.13}$$

with $\nu_\Gamma(du) = e^{-u/2}(1 - e^{-2u})^{-1} du$. Moreover, if

$$d\mu_K^+(x) = 2 \mathbf{1}_{[0,\infty)}(x) d\mu_K(x), \tag{24.14}$$

then μ_K^+ is a probability measure and

$$\mathrm{Var}_{\mu_K}(r) = \mathrm{Var}_{\mu_K^+}(r). \tag{24.15}$$

Proof. Orient the prime displacement as

$$\int_{\mathbb{R}} K(x) K(x + \log d) |r(x) - r(x + \log d)|^2 \, dx.$$

The two same-sign rays $x \geq 0$ and $x \leq -\log d$ agree by evenness and give twice the positive half-line integral. The interval $(-\log d, 0)$, after reflection, gives the second line of (24.12). From (24.8),

$$k_m(x + \log d) = \sqrt{d}\, k_{dm}(x). \tag{24.16}$$

Therefore

$$\frac{\Lambda(d)}{\sqrt{d}} K(x)K(x + \log d) = \Lambda(d)K(x) \sum_{m \geq 1} k_{dm}(x).$$

Summing in d and grouping $\ell = dm$ proves the first line of (24.12). The same ray/crossing split gives (24.13). Finally, K, h and r are even, so conditioning μ_K on the positive half-line changes neither moment of r, proving (24.15). □

Proposition 24.2 is not another RH-equivalent criterion. It is an exact arithmetic coordinate for the already-known criterion: a half-line chain with theta–divisor outward edges, central reflection edges and Gamma continuum edges. The outer factor $K(x)$ retains all theta indices, including those which are nondivisible for a given displacement; the crossing channel is kept rather than estimated away.

Open problem 24.3 (All-source sharp gate)**.** Prove, with all terms in (24.2) retained,

$$\boxed{\mathscr{E}_K(r) \geq \frac{1}{2} \operatorname{Var}_{\mu_K}(r) \qquad (r \in C^\infty_{c,\mathrm{even}}(\mathbb{R})).} \tag{24.17}$$

Equivalently, prove

$$\sigma(\mathcal{L}_K|_{\mathcal{H}_{0,+}}) \subset [1/2, \infty). \tag{24.18}$$

A constructive closure of (24.17) may take either of two equivalent operator forms. The strongest is an arithmetic factorization

$$\mathscr{E}_K(r) - \frac{1}{2} \operatorname{Var}_{\mu_K}(r) = \|\mathcal{B}r\|^2_{\mathcal{Y}} \tag{24.19}$$

on the quotient by the threshold radical, where $\mathcal{B}$ must be built from the complete divisor–theta coupling rather than from generic jump positivity. A second route is to prove (23.12), choose a reference operator with threshold 1/2, and exclude its subthreshold Birman–Schwinger spectrum by an all-source norm bound. In either route, the equality modes of Proposition 23.3 determine the sharp normalization and forbid any strict global improvement of the constant.

More explicitly, put

$$\mathcal{R}_K = \overline{\operatorname{span}} \left\{ K^{(2m)}/K - 4^{-m} : m \geq 1 \right\} \subset \mathcal{H}_{0,+}. \tag{24.20}$$

Proposition 23.3 gives $\mathcal{L}_K|_{\mathcal{R}_K} = \frac{1}{2}I$. The unresolved coercive statement is therefore the same sharp floor on $\mathcal{R}_K^\perp$, with the three exact channels (24.12)–(24.13) kept coupled.

25. Conclusion

Starting with the finite self-adjoint quotients of CCM and the continuous localized perspective of Suzuki, this paper constructs a single chain through the two limiting parameters. At fixed physical cutoff, the quotients possess a canonical metric-compatible transport; its defect is a rank-two source expression plus its Loewner metric correction. This yields the resolvent-shell identities, cofinal fixed-cutoff summability and the precise adjacent-cutoff obstruction.

In the outer limit, logarithmic curvature is shown to be an affine-rigid divisor observable. The moving co-Poisson construction keeps primes, prime powers, Gamma and pole coupled; the centered jump identity restores the Fourier center lost by translated mean-square estimates; and exact PNT compensation isolates the signed discrepancy channel. These results identify what the finite operator convergence program must still prove: the global branch selector together with the moving/model identification.

The full Riemann kernel provides a second, cutoff-free endpoint of the same development. Its smooth truncations are operator quasimodes, and the exact radical identity converts the completed Weil form into

$$QW(Kr, Kr) = \mathscr{E}_K(r) - \frac{1}{2}\operatorname{Var}_{\mu_K}(r).$$

The resulting RH equivalence is a ground-state normalization of the classical Weil criterion, not a new criterion and not progress by itself toward proving its sign. Its role in this paper is to expose the literal prime-power and theta channels of the unresolved global branch. Within that normalization, the polar term gives the exact constant $1/2$, and the complete prime-power–Gamma measure remains visible in $\mathscr{E}_K$. Corollary 23.2 therefore identifies direct Weil positivity, hence RH, with the spectral floor

$$\inf \sigma(\mathcal{L}_K|_{\mathcal{H}_{0,+}}) \geq \frac{1}{2}.$$

The threshold is sharp and highly degenerate: Proposition 23.3 places an infinite family of Riemann-radical modes exactly at $1/2$. Theorem 24.1 also proves that every literal von Mangoldt atom is load-bearing. The exact theta dilation cannot be weakened to its divisible part, because the nondivisible theta indices and the central crossing region carry a strictly positive part of the threshold energy. Thus finite-prime certificates, positive-tail completion, generic Picone identities and termwise dilation estimates cannot establish the sharp floor.

There are consequently two related but distinct unresolved statements. For convergence of the particular CCM–Suzuki spectral approximants, the remaining conditions are the signed branch selector and moving/model identification. For a direct proof of RH in the full-kernel coordinate, the single remaining theorem is Problem 24.3: preserve all prime atoms, divisible and nondivisible theta indices, the central crossing, Gamma and the polar variance, and prove that no spectrum lies below $1/2$. Proving first that the essential spectrum has floor $1/2$ would reduce the final step to excluding discrete subthreshold states. Neither the sharp inequality nor the full operator convergence is claimed here.

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